3

In this chapter, we will learn about

- some basic ideas about forces
- inertia and Newton's first law of motion
- net force and Newton's second law of motion
- weight and the feeling of weight
- friction and fluid resistance
- Newton's third law of motion

Force and motion (I)

3.1 Force

In the last two chapters, we learned how to describe motion. In this chapter, we will investigate how a motion is produced.

Flipped classroom
Watch a video and answer the questions.

DSE goal
Use newton as a unit of force.

1 What are forces?

We all know that motion is related to **forces**. In junior science, we have learned that forces have the following properties:

1 A force can speed up or slow down an object.

2 A force can change the moving direction of an object.

3 A force can change the shape of an object.

In general, a force is a push or pull in a particular direction. It is a **vector** quantity (like displacement and velocity). An arrow is used to represent a force in drawings to indicate its direction and size (Fig 3.1a).

You will learn more about this in Ch 3.7. ▶

Furthermore, **forces exist in pairs**. When you are exerting a force on a suitcase, the suitcase is also exerting a force on you (Fig 3.1b).

(i) Pulling force on suitcase.

(ii) Kicking force on football.

Fig 3.1a *Examples of applying force.*

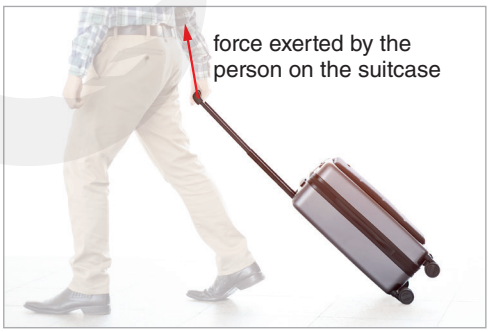


Fig 3.1b *Forces exist in pairs.*

force 力

The unit of force is the **newton**, written as **N**. Forces can be measured by ***spring balances*** or ***force sensors*** (Fig 3.1c). Figure 3.1d shows some typical magnitudes of forces.

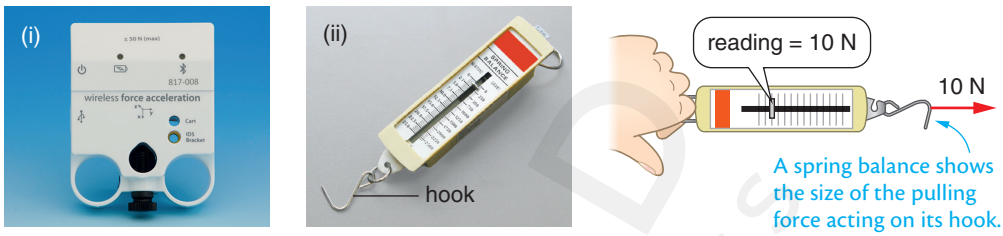


Fig 3.1c *(i) A force sensor, (ii) a spring balance.*

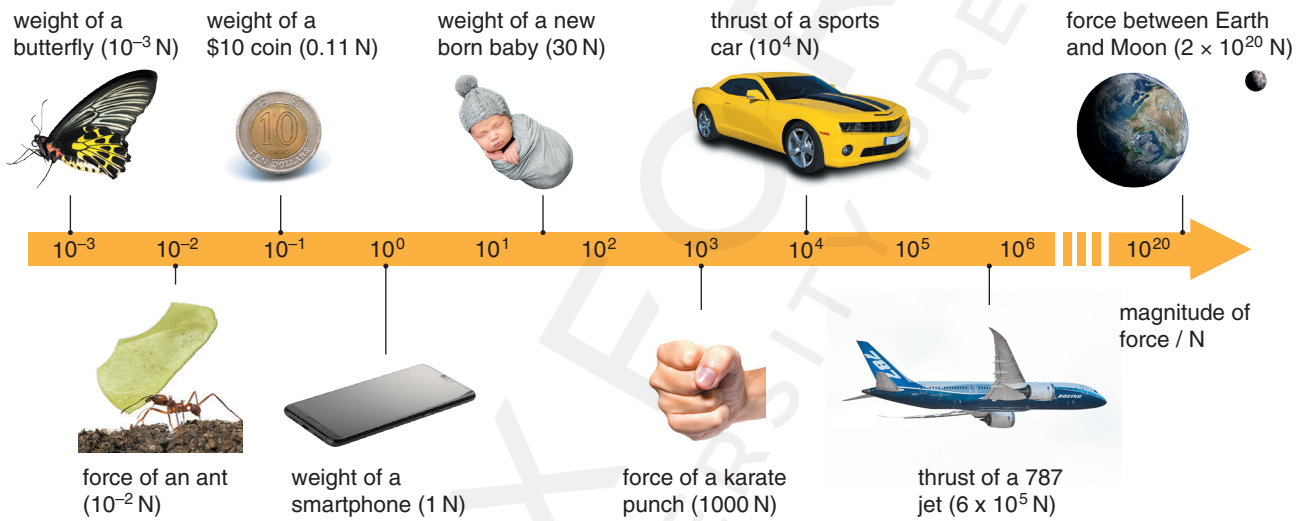


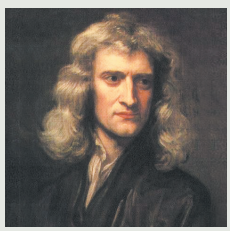
Fig 3.1d *Some typical magnitudes of forces in daily life.*

Know more

Isaac Newton (1642–1727)

Isaac Newton did a lot of important work on mechanics that changed people's understanding of nature. His three laws of motion are the foundation of classical mechanics. The unit of force is named after him in honour of this great scientist.

OUP eResources



DSE goal

Understand friction as a force opposing motion or tendency of motion.

2 Different types of forces

Force is everywhere around us. They can be classified into *contact forces* and *non-contact forces*.

a Contact force

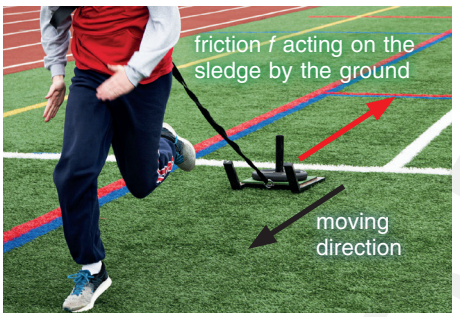
Both pulling a suitcase and kicking a football are examples of exerting contact force. There are also other types of contact force in daily life.

newton (N) 牛頓　spring balance 彈簧秤　force sensor 力感應器　Isaac Newton 牛頓
contact force 接觸力　non-contact force 非接觸力

i Friction

Friction *f* is a force which arises when two objects are sliding or tend to slide over one another. The direction of friction is **always opposite** to the relative motion or the tendency of relative motion between two surfaces (Fig 3.1e).

(i) A sledge sliding on the ground.



(ii) A boy sitting on a slope.

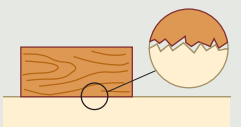
Fig 3.1e *Friction always opposes relative motion or the tendency of relative motion between surfaces.*

Most of the surfaces in daily life are **rough** (will exert friction on objects). However, for the sake of convenience in calculation, we sometimes assume a surface to be **smooth** (will not exert friction on objects).

Know more

The origin of friction

The existence of friction may be explained by tiny bumps on surfaces. When two surfaces are in contact, these tiny bumps catch each other and cause friction.



ii Normal force

Normal means perpendicular.

Normal force *R* is a force acting **perpendicularly** on an object **by the surface in contact** with it (Fig 3.1f). You do not fall to the centre of the Earth because you are supported by the normal force from the ground.

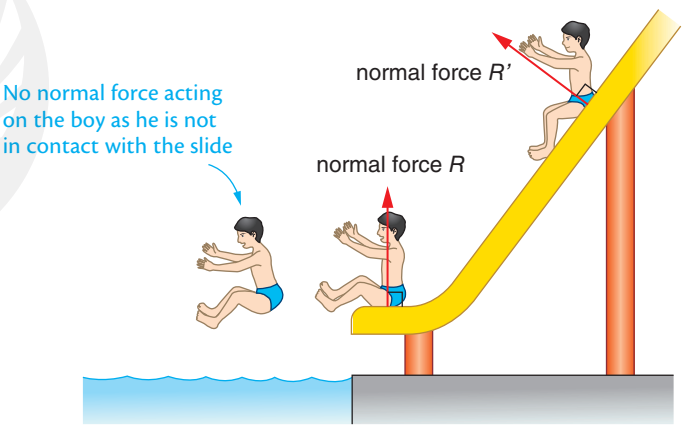


Fig 3.1f *The slide exerts a normal force on the boy when they are in contact.*

friction 摩擦力　　normal force 法向力

iii Tension

When a string is **stretched**, the force called **tension** *T* is developed along the string. For example, in a tug of war, the players experience tension when the rope is being pulled at both ends. Tension always **pulls** objects **along** the string (Fig 3.1g). If a light inextensible string remains straight, its tension has the same magnitude at every point.

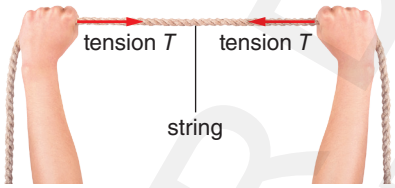


Fig 3.1g *Tension in a straight string.*

b Non-contact force

i Weight

The **weight** *W* of an object is the *gravitational force* acting on the object by the Earth. It always points to the centre of the Earth. Even when the object is not in contact with the Earth, there is also weight acting on the object (Fig 3.1h).

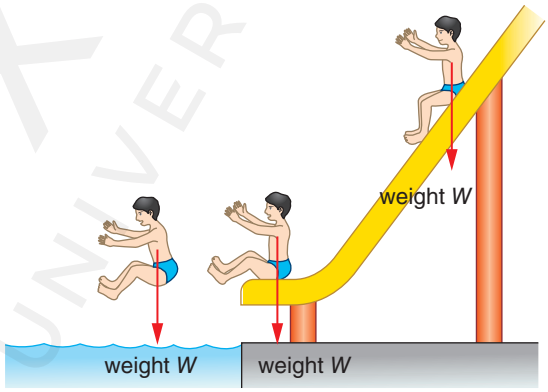


Fig 3.1h *Weight always acts on the boy.*

→ This video demonstrates the magnetic force.

Video

Show the video and ask Ss what they would feel if they pull the globe downwards a little bit.

ii Electrostatic and magnetic force

An *electrostatic force* exists between electric charges (Fig 3.1i) while a *magnetic force* exists between magnets (Fig 3.1j). Both kinds of force can act at a distance. They can be attractive or repulsive.

The electrostatic forces and the magnetic forces exist in pairs and are 'action-and-reaction pairs'. You will learn about this in Ch 3.7. ▶

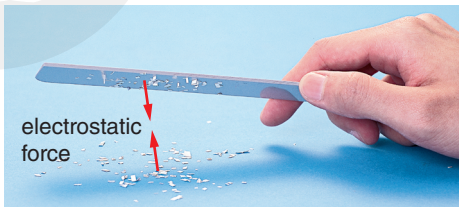


Fig 3.1i *A charged plastic ruler attracts small bits of paper without touching them.*

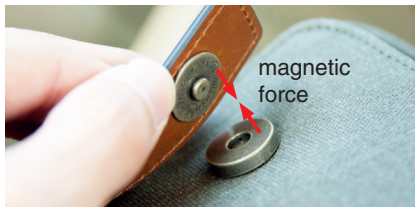


Fig 3.1j *We can feel the attraction between magnets at a distance.*

tension 張力　weight 重量　gravitational force 重力 / 引力　electrostatic force 靜電力
magnetic force 磁力

DSE goal

Use free-body diagrams to show the forces acting on objects.

DSE exam

15(1B)Q4(c), 19(1A)Q8

→ Teaching notes and exercises are available on OUP web.

Skill *Drawing*

Drawing free-body diagrams

① Draw only one object. Do **not** draw the other objects that interact with it.

② For a complicated shape, simply draw it as a rectangle.

③ Identify all the forces **acting on the object**. Ignore the forces acted by it on other objects.

④ Draw each force as an arrow in the correct direction. It is a good practice to draw a longer arrow to represent a larger force (but not necessarily in scale) and draw the arrow from the point where the force acts on.

⑤ Label the arrows.

3 Free-body diagrams

A **free-body diagram** is used to display all the forces acting on an object.

In Figure 3.1k, when the sledge is pulled by the huskies, there are four forces acting on the sledge, namely tension, normal force, weight and friction. From that, we can draw the free-body diagram for the sledge (Fig 3.1l).

Fig 3.1k *A sledge pulled by huskies.*

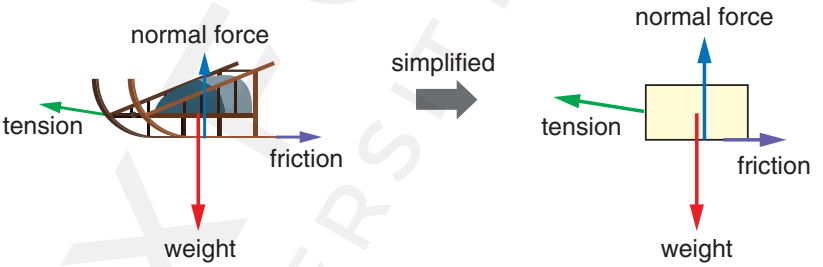
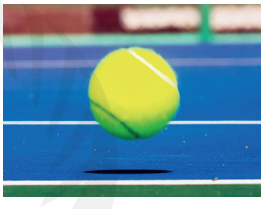


Fig 3.1l *Free-body diagram for the sledge.*

Example 1 Drawing free-body diagrams

Draw a free-body diagram for each of the following objects.

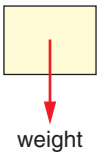
(a) A ball falling freely



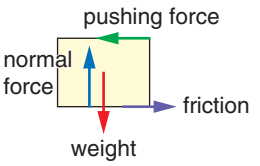
(b) A box being pushed to the left

Solution

(a)



(b)



Weight always acts vertically downwards for objects on the Earth. ▶

free-body diagram 隔離體圖 / 自由體圖

DSE goal

Determine the net force acting on objects.

The addition of forces in a plane will be discussed in Ch 4.1. ▶

4 Net force

When two or more forces are acting on an object, all the forces can be added up and the result is the **net force**, which determines how the motion of an object changes. Table 3.1a shows the addition of parallel forces.

Separate forces	Net force
+ → 2 N, 2 N	net force = (+2) + (+2) = 4 N → 4 N
2 N ← 2 N	net force = (+2) + (–2) = 0
4 N ← 2 N	net force = (+2) + (–4) = –2 N ← 2 N

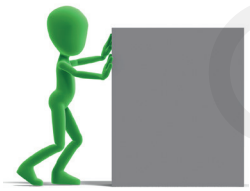
Table 3.1a *Addition of parallel forces, taking the direction to the right as positive.*

Quick check 1

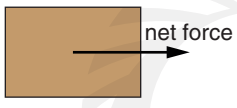
→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

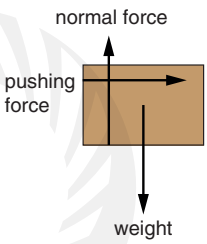
1 A box on a **smooth** floor is being pushed. Which of the following gives the correct free-body diagram for the box?



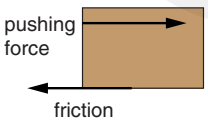
A



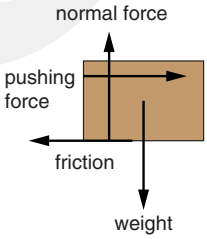
B



C



D

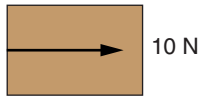


2 Which of the following statements about different types of forces is/are correct? Put a tick in the corresponding box(es).

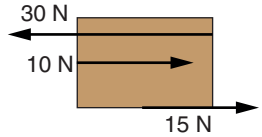
- ☐ Tension in a string always acts along the string.
- ☐ Weight acts on an object only when the object is in contact with the ground.
- ☐ Normal force always points upwards.
- ☐ Friction arises only when an object moves.

3 Find the net force acting on the block in each of the following cases.

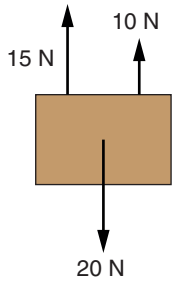
(a)



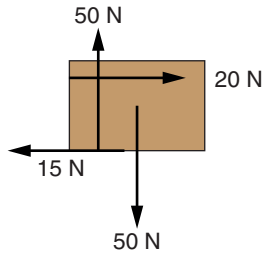
(b)



(c)



(d)

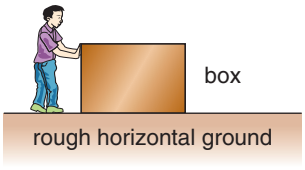


net force 淨力

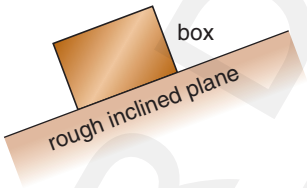
Learn by practice 1

1 Draw a free-body diagram for the box in each of the following cases.

(a) Being pushed on rough horizontal ground



(b) Resting on a rough inclined plane



(c) Free falling

Exercise 3.1

1 Which of the following statements about force is **incorrect**?

A Force is a vector.

B Forces always exist in pairs.

C An object can exert a force on another object only if the two objects are in contact.

D The unit of force is newton.

2 Anson and Benny pull a string as shown (Fig a).

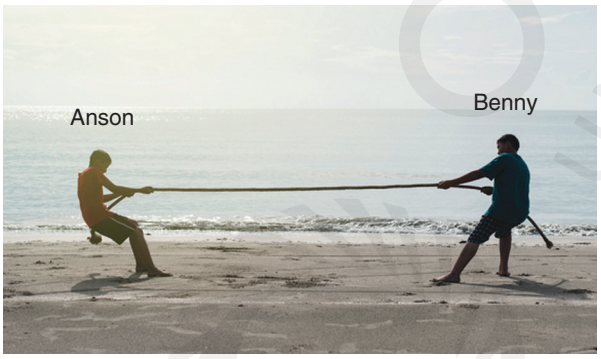


Fig a

(a) Draw free-body diagrams for Anson and Benny separately.

(b) Does the tension acting on Anson have the same magnitude as that acting on Benny?

3 A box is acted on by two forces: its weight and a force F. The weight of the box is 10 N and the net force acting on the box is 12 N upwards (Fig b). What are the magnitude and direction of F?

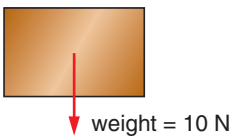


Fig b

*4 A figure is hung from a string (Fig c).

Fig c

(a) Draw a free-body diagram for the figure.

(b) The magnitude of the tension in the string is 13 N and the weight of the figure is 10 N. Find the magnitude and direction of the net force acting on the figure.

*5 Objects X and Y are hung from strings as shown (Fig d). Draw free-body diagrams for objects X and Y separately.

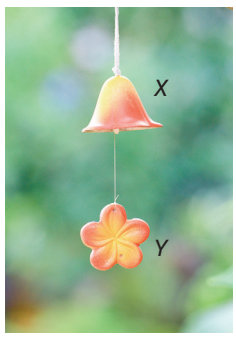


Fig d

3.2 Newton's first law of motion

1 Force and inertia

DSE goal
Describe the meaning of inertia.

We need to provide a force to make an object start moving from rest. Then, is a force necessary to keep the object moving?

Fig 3.2a *Force seems to be needed to keep objects moving in daily life.*

In most daily life experience, we do need to provide a force to keep an object moving. However, there are also situations in which objects can move a long distance without a force that keeps pushing them (Fig 3.2b). These examples show that force is not a must to keep an object moving.

Fig 3.2b *Some objects go a long way without a force that keeps pushing them.*

Know more

Aristotle and Galileo

The Greek philosopher *Aristotle* thought that a force was needed to keep an object moving. This was believed to be true until it was challenged by Galileo.

▸ p.152

For a very long time in the past, people believed that a force was needed to keep something moving. When the object was 'using up the force', it slowed down. Towards the end of the 16th century, Galileo started to question this belief.

Based on a series of careful observations, experiments and reasoning, Galileo proposed that a moving object slowed down because there was friction that opposed its motion. In the absence of friction, he argued, a moving object would move on forever at a constant speed along a straight line (i.e. constant velocity).

More precisely, in the motion on an air track, the friction is so small that it can be neglected. ▶ Today, we can demonstrate frictionless motion using an air track.

Aristotle 亞里士多德

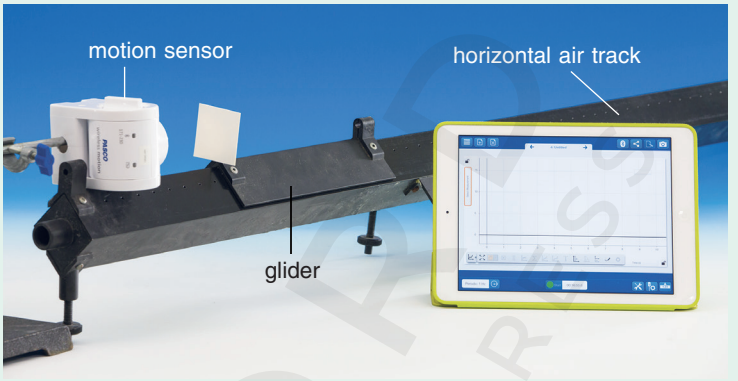
Video

→ This video demonstrates Expt 3a.

→ This is a Video Quiz about this experiment.

Experiment 3a / Motion with and without friction

1 Set the apparatus as shown.



2 Without turning on the pump, give a soft push to the glider from a position close to the motion sensor.

3 Turn on the air pump. Give a soft push to the glider from the same position.

An air track provides a layer of air for the glider to move on. This reduces the friction acting on the glider. ▶

Discussion

Compare the motion of the glider in steps 2 and 3.

Figure 3.2c shows an *s*–*t* graph obtained in step 3 in Experiment 3a. The result verifies that when there is no friction, an object moves at a constant velocity. No force is needed to keep it moving.

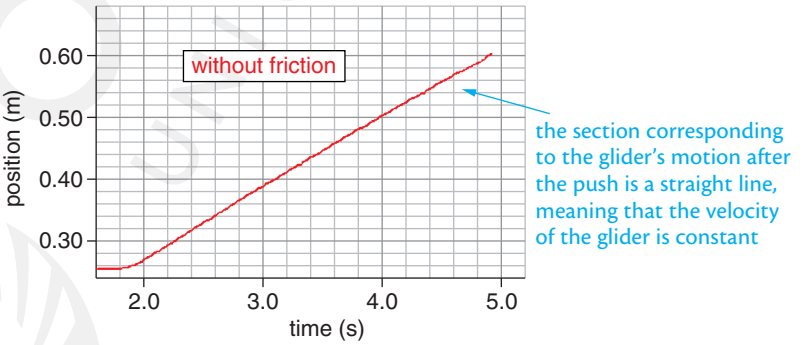


Fig 3.2c *An s–t graph of the glider moving without friction acting on it.*

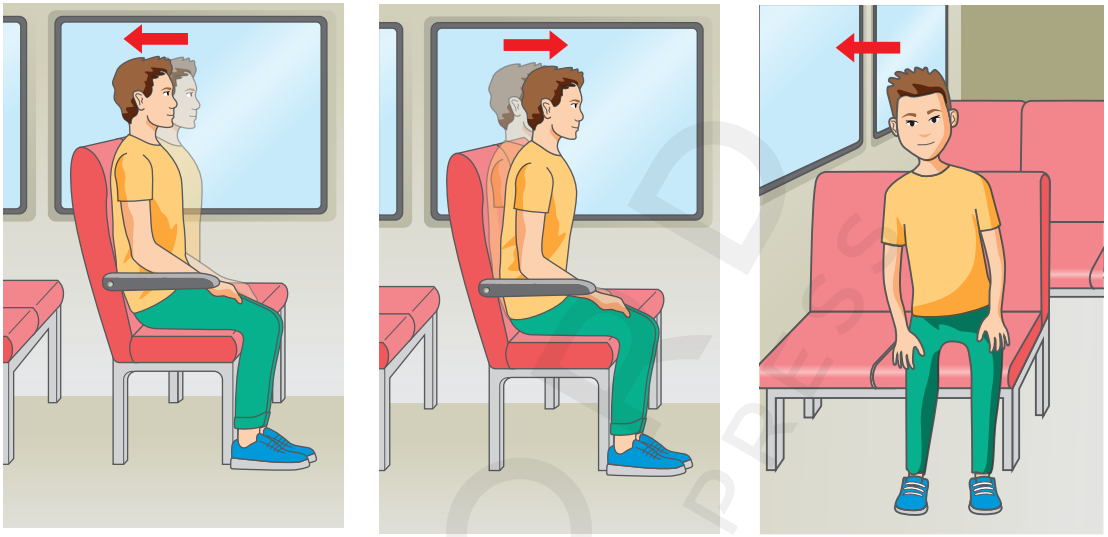
On the other hand, a stationary object would remain at rest if it is undisturbed. These give rise to the concept of **inertia**.

Inertia is the tendency of an object to maintain its state of rest or of constant velocity.

This is known as *Galileo's law of inertia*.

inertia 慣性　　Galileo's law of inertia 伽利略慣性定律

It is not difficult to find everyday examples of inertia. Figure 3.2d shows three different situations that a person may face in a bus.



When the bus starts moving, the person seems to move backwards because he has the tendency to **remain at rest**.

When the bus slows down, the person is 'thrown forwards' because he has the tendency to **keep moving** forwards **at the original speed**.

When the bus makes a sharp turn to the left, the person is 'thrown to the right' because he has the tendency to **keep moving in the original direction**.

Fig 3.2d *Examples of inertia of a person in a bus.*

Some more tricks of inertia:

Video

→ The video 'Coin tower' demonstrates a trick of inertia.

→ The video 'Saving the banknote' demonstrates a trick of inertia.

→ This video demonstrates the Hands-on activity.

Video

▶ The trick below is related to inertia.

Hands-on activity

Inertia trick

What you need

What to do

Put the coin and the card on your finger as shown. Flick the card away quickly.

What you find

What happens to the coin?

DSE goal

Describe the relationship of inertia to mass.

2 Inertia and mass

The inertia of an object is related to its **mass**. Their relationship can be shown in the following activity (Fig 3.2e). When the cloth is pulled very quickly, bottle *A*, which is fully filled, remains standing while the empty bottle *B* falls down. This shows that an object with a greater mass has a higher tendency to maintain its state of rest, i.e. has a larger inertia.

Video

→ The video demonstrates the relationship between mass and inertia.

Fig 3.2e *Demonstrating the relationship between mass and inertia.*

Mass is a measure of an object's inertia.

The SI unit of mass is the **kilogram**, written as **kg**.

The greater the mass, the greater the inertia, the more difficult it is to change the velocity. Figure 3.2f shows more examples.

(i) It is much harder to move the heavier punching bag than the lighter one.

(ii) A massive cruise ship finds it much harder to change its velocity than a small boat.

Fig 3.2f *The greater the mass, the greater the inertia.*

mass 質量　　kilogram (kg) 公斤 / 千克

Google Forms

Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 *True or false*:

(a) A moving object will gradually stop because the force in it is gradually used up. (T/F)

(b) A moving object will gradually stop because of inertia. (T/F)

(c) Force is not a must to keep an object moving. (T/F)

2 A moving object with a (larger size / larger mass / smaller size / smaller mass) is more difficult to (be kept moving / be stopped) as it has a larger inertia.

3 A man is in a train heading towards the left. When the train is moving steadily, the man looks as shown in Figure a. Which figure shows that the train is braking?

Fig a

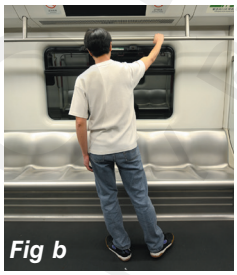


Fig b

Fig c

DSE goal

State Newton's first law of motion and use it to explain situations in which objects are at rest or in uniform motion.

DSE exam

20(1A)Q5

3 Newton's first law of motion

The pioneering work of Galileo on motion was further developed by Newton. In 1687, Newton published his three laws of motion. These laws are among the most basic laws in physics and are still used today to make calculations and predictions on moving objects.

Newton's first law of motion states that:

> **Every object remains in a state of rest or uniform motion (i.e. constant velocity) unless it is acted on by a net force.**

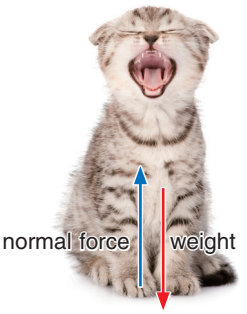
net force = 0 ↔ change in velocity = 0 ↔ object remains at rest / object remains in uniform motion

In simple terms, when the net force acting on an object is zero, an object at rest will stay at rest, and an object in uniform motion will keep moving in a straight line with a uniform velocity.

This law is similar to Galileo's law of inertia except that Newton gave a definite meaning to force: a force changes the state of rest or uniform motion of an object. That is, a force is not required to keep an object moving. Instead, it changes the state of motion of an object.

More precisely, the force here refers to the net force. ▶

We can explain many situations using Newton's first law of motion. The following are some examples (Fig 3.2g).



A cat is sitting still on the ground. There are two forces acting on it, namely its weight and the normal force by the ground. These two forces cancel each other out and there is no net force acting on the cat. As a result, it remains at rest.

A girl is ice-skating. The friction is very small and is negligible. Therefore, only two forces, the weight and the normal force by the ground, act on the girl and the net force is zero. The girl, initially moving, keeps moving with a constant velocity (in uniform motion).

The cart will slow down and eventually stop. ▶

A boy pushes a cart on level ground. There are two forces acting on the cart along the horizontal direction, namely the pushing force and friction.

- If these two forces cancel each other out, the cart will move at a constant velocity.
- If the boy lets the cart go, the net force is no longer zero and the cart will change its state of motion.

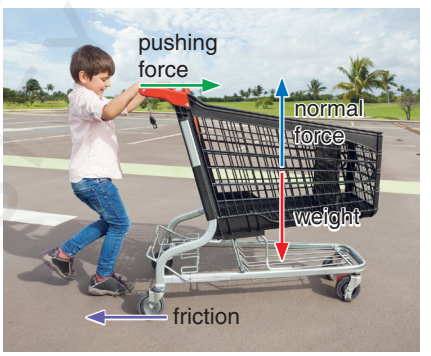


Fig 3.2g *Explaining motion using Newton's first law of motion.*

Example 2 Man inside a lift

A man is standing in a lift which is moving downwards at a constant velocity.

(a) Name the forces acting on the man.

(b) Is there any net force acting on the man? Explain your answer.

Solution

(a) Weight and normal force acting by the floor of the lift

(b) Since he is moving at a constant velocity with the lift, by Newton's first law of motion, there is no net force acting on him.

STSE

Seat belts and headrests in car safety

If a car travelling at a high speed suddenly stops, the people inside will continue to move forwards at a high speed due to inertia. They may hit the windscreen and be thrown out of the car. This would result in serious injuries or even death.

If they wear seat belts, the seat belts will exert forces on them to stop them, thus reducing injury.

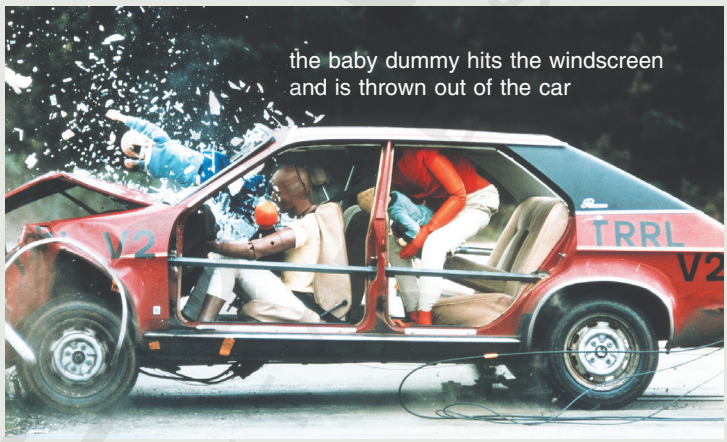
On the other hand, if a stationary car is bumped at the back in a crash, people in the car will be pushed forwards by the seats but their heads tend to remain at rest. Headrests are installed to prevent serious head and neck injuries.

seat belt

Video

headrest

Video



Google Forms

Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 Is there (i) any force, (ii) any net force acting on each of the following objects? Put a tick in the box if the force exists.

	(i)	(ii)
(a) A car at rest	☐	☐
(b) A car speeding up	☐	☐
(c) A car turning a corner with constant speed	☐	☐
(d) A car moving along a straight line with constant speed	☐	☐
(e) A car moving in a circle with constant speed	☐	☐
(f) A spaceship travelling with constant velocity in outer space	☐	☐
(g) A skydiver falling with constant velocity	☐	☐

2 A man is pushing a box on level ground (Fig a). What is the relationship between his pushing force P and the friction f acting on the box in the cases below?

Fig a

(a) The box is not moving.

A $P < f$ B $P = f$ C $P > f$

(b) The box is moving with a constant velocity.

A $P < f$ B $P = f$ C $P > f$

Exercise 3.2

1 Cat *X* runs forwards at a constant velocity of 10 m s^{-1}. Cat *Y* at the back runs at a constant velocity of 12 m s^{-1} (Fig a).

Fig a

Does a net force act on any of the cats?

A *X* only
B *Y* only
C *X* and *Y*
D Neither of them

★**2** Which of the following statements is/are correct?

(1) When moving at the same speed, a bowling ball is harder to stop than a ping-pong ball.
(2) Objects always have the same inertia when they are at rest.
(3) When a bus accelerates forwards suddenly, passengers inside lean backwards because they are acted on by friction.

A (1) only
B (3) only
C (2) and (3) only
D (1), (2) and (3)

3 A rabbit remains still on the ground (Fig b).

Fig b

(a) Name the forces acting on it.
(b) Why can it remain still when it is acted on by the forces in (a)?

4 Spaceship Voyager 2 (Fig c) was launched in 1977. It is now over 19 billion kilometres from the Earth. It can travel continuously in space even though its rockets are shut down. Explain why.

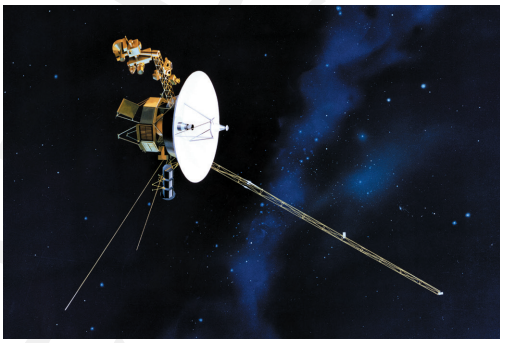


Fig c

★**5** Bottle *A* filled with water and empty bottle *B* are resting on a plastic sheet (Fig d). Explain what happens to the bottles when the plastic sheet is pulled away quickly.

Fig d

★**6** Explain, in terms of inertia, why we should hold the handrails on a bus (Fig e).

Fig e

3.3 Newton's second law of motion

1 Net force, acceleration and mass

DSE goal

Describe the effect of net force on speed and direction of motion of an object.

State Newton's second law of motion.

Verify $F = ma$ experimentally.

In the last section, we learned that a net force changes the state of rest or uniform motion of an object. This means that the velocity of the object will change. In other words, it will have an acceleration.

net force ➡ velocity changes (with acceleration)

On the other hand, the concept of inertia tells us that the greater the mass of an object, the more difficult it is to change its velocity (Fig 3.3a). Hence acceleration is also related to mass.

Fig 3.3a *It is much easier to change the velocity of a toy car than a real car.*

Newton's second law of motion gives a quantitative relationship between the net force acting on an object, the mass and the acceleration of the object (Fig 3.3b):

> **The acceleration *a* of an object is directly proportional to, and in the same direction as, the net force F_{net} acting on it, and inversely proportional to the mass *m* of the object.**

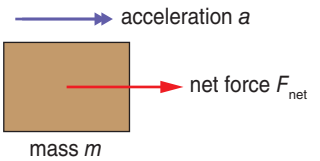


Fig 3.3b *Acceleration a, net force F_{net} and mass m.*

Mathematically, this relationship can be expressed as $a \propto \frac{F_{net}}{m}$. We can verify this relationship in Experiment 3b.

→ The simulation 'Acceleration and net force' simulates Part I of Expt 3b.

→ The video 'Expt 3b - Relationship between acceleration, net force and mass (Part I)' demonstrates Part I of Expt 3b.

Simulation Video

→ The simulation 'Acceleration and mass' simulates Part II of Expt 3b.

→ The video 'Expt 3b - Relationship between acceleration, net force and mass (Part II)' demonstrates Part II of Expt 3b.

The elastic thread should be stretched by the same amount to ensure the pulling force is constant. ▶

Skill *Experiment*

Friction-compensated runway

A friction-compensated runway is one where the effect of friction is compensated for by a slope. One end of the runway is slightly raised. A trolley can remain at rest on the runway. When it is given a soft push down the runway, it runs down at a constant speed (can check by looking at the *v*–*t* graph obtained). Using a friction-compensated runway ensures that the pulling force on the trolley is the net force.

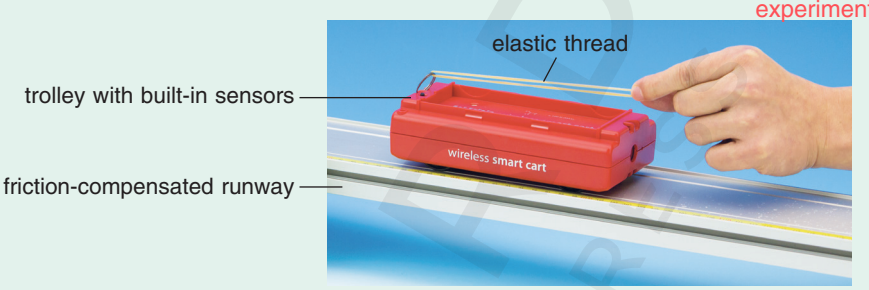
→ Teaching notes and exercises are available on OUP web.

Experiment 3b / Relationship between acceleration, net force and mass

Threads of weak elasticity are recommended. Otherwise, the trolley will be moving so fast that it is not easy to maintain the same tension throughout the experiment.

Part I How does net force affect acceleration?

1 Set up the apparatus as shown. Adjust the runway for friction compensation.



2 Start data-logging. Move the trolley by pulling the elastic thread. The elastic thread should be stretched by the same amount all the time.

3 Find the acceleration a of the trolley from the v–t graph obtained.

4 Repeat steps 2 and 3 with 2, 3 and 4 elastic threads stretched by the same amount.

5 Plot a graph of a against the number of elastic threads ($\propto$ net force F).

Emphasize that the runway is smooth (frictionless) or friction-compensated, so that F is the net force.

Part II How does mass affect acceleration?

1 Repeat steps 2 and 3 in Part I by changing the mass m of the trolley.

2 Plot a graph of a against $\frac{1}{m}$.

Ss may be puzzled to see that a is plotted against $\frac{1}{m}$. Ts may plot the data with a against m first and explain that it is not easy to deduce the relation between them from this graph.

Discussion

1 What can you conclude from the graph of a against the number of elastic threads?

2 What can you conclude from the graph of a against $\frac{1}{m}$?

Figures 3.3c and 3.3d show the results of the experiment.

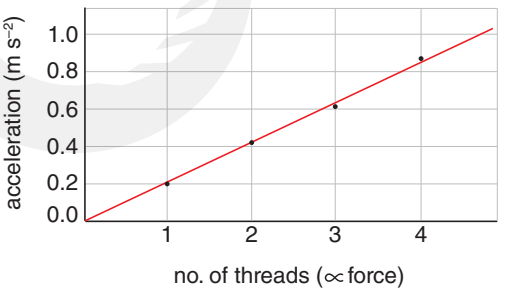


Fig 3.3c Graph of a against number of threads.

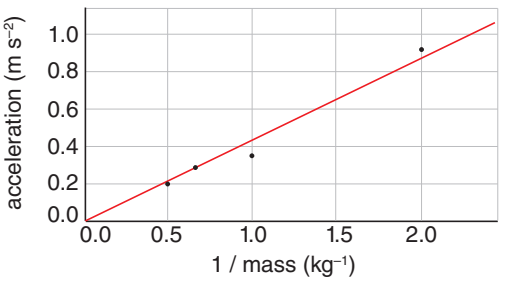


Fig 3.3d Graph of a against $\frac{1}{m}$.

From Figure 3.3c,

$a \propto$ number of elastic threads

$\Rightarrow \; a \propto F_{net}$ (constant m) .....(1)

From Figure 3.3d,

$$a \propto \frac{1}{m} \text{ (constant } F_{net}) \quad \text{.....(2)}$$

A relationship satisfying both (1) and (2) is

$$a \propto \frac{F_{net}}{m}$$

This verifies Newton's second law of motion.

Know more

Direct proportion and inverse proportion

If y is directly proportional to x, we can write $y \propto x$ or $y = kx$. The graph of y against x (Fig a) is a straight line through the origin with slope k.

If y is inversely proportional to x, we can write $y \propto \frac{1}{x}$ or $y = \frac{k}{x}$. The graph of y against $\frac{1}{x}$ (Fig b) is a straight line through the origin with slope k.

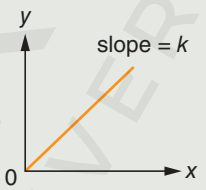


Fig a $y \propto x$.

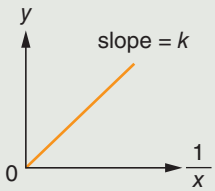


Fig b $y \propto \frac{1}{x}$.

The relationship $a \propto \frac{F_{net}}{m}$ can be written as

$$F_{net} \propto ma \quad \text{or} \quad F_{net} = k \times ma$$

where k is a constant.

The SI unit of force is the **newton**, written as N. We define this unit to make $k = 1$. That is, one newton of net force produces an acceleration of 1 m s^{-2} on a mass of 1 kg. Hence Newton's second law of motion is commonly written as

1 N = 1 kg m s^{-2} ▶

net force (N) mass (kg) acceleration (m s^{-2})

$$F_{net} = ma$$

Note the following:

1 F_{net} refers to the net force acting on the object. For example, in Figure 3.3e, the pulling force P is not equal to ma because P is not the net force.

2 The net force F_{net} and the acceleration a are always in the same direction. Recall that the acceleration alone does not tell us the direction of motion, or whether the object is speeding up or slowing down. Similarly, the net force alone does not tell us such information (Fig 3.3f).

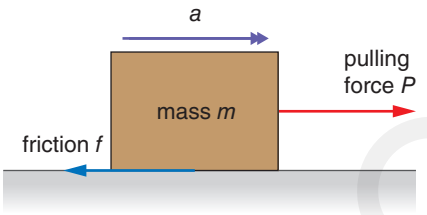


Fig 3.3e *The net force acting on the box is P – f, not P.*

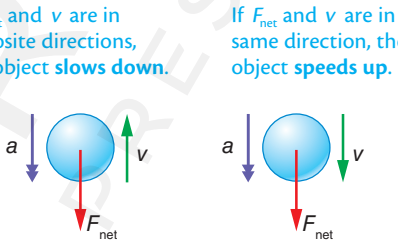


Fig 3.3f *To determine whether an object speeds up or slows down when a net force acts on it.*

Quick check 4

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 A spaceship is travelling forwards in outer space (Fig a). It is moving with a constant velocity v.

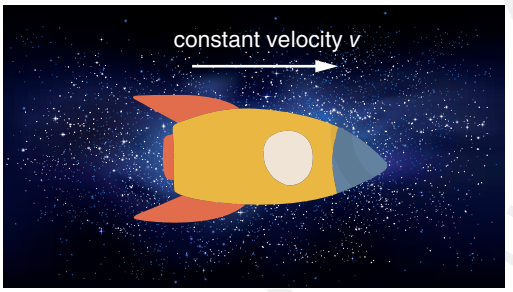


Fig a

(a) Is there a net force acting on the spaceship?
A Yes B No

(b) The spaceship then fires its rockets so that it is acted on by a forward force. What happens to the spaceship?
A It travels with a constant velocity higher than v.
B It moves faster and faster.

(c) What happens to the spaceship after the rockets are shut down?
A It travels with a constant velocity higher than v.
B It slows down to a stop.

2 A man throws a ball straight up a short distance (Fig b). The ball slows down and comes to a stop momentarily at the highest point before falling down.

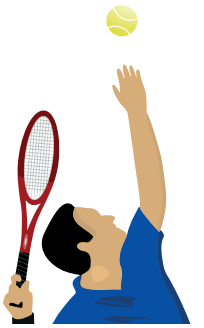
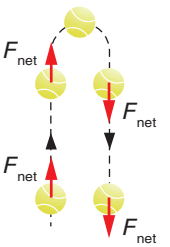


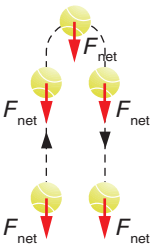
Fig b

(a) What are the directions of the net force F_{net} acting on the ball at different points?

A



B



(b) Why does the ball slow down as it goes up?
A The force given to the ball is gradually used up as the ball goes up. The ball stops when all the force is used up.
B When the ball goes up, the net force and its velocity are in opposite directions, so it slows down.

DSE goal

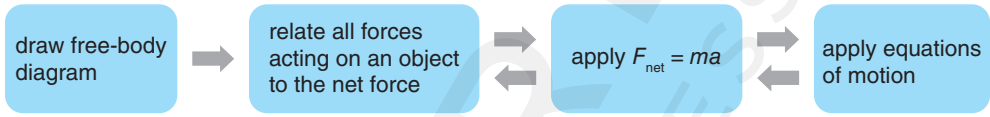
Apply Newton's second law of motion to solve problems involving motion in one dimension.

DSE exam

13(1B)Q6(a)(i), 15(1A)Q5, 23(1B)Q3(c)

2 Solving problems using Newton's second law of motion

Newton's second law can be used to solve a wide range of problems. Before we can apply $F_{net} = ma$, we should first identify all the forces acting on the object by drawing a **free-body diagram** and then use it to find F_{net}. You may refer to the following flow chart when solving related problems.

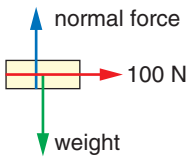


For an object moving in a straight line, forces perpendicular to its moving direction always cancel each other out and can be ignored. The net force, if any, must point along its moving direction.

Example 3 Acceleration of a puck

Tom hits an ice hockey puck to the right with a horizontal force of 100 N. The mass of the puck is 0.17 kg. The friction acting on the puck is negligible. Find the acceleration of the puck.

Solution

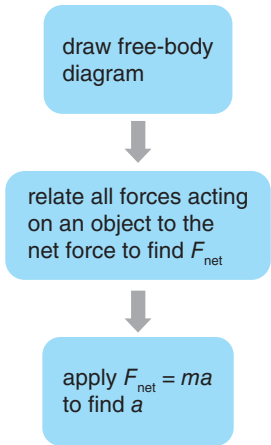


F_{net} = 100 N (rightwards)

By $F_{net} = ma$,

$$a = \frac{F_{net}}{m} = \frac{100}{0.17} = 588 \text{ m s}^{-2} \text{ (rightwards)}$$

Flow chart for this example:



Tip

F_{net} acts along the moving direction. We need to consider only the force(s) acting along this direction when applying $F_{net} = ma$. The weight and the normal force cancel each other out.

Example 4 | Water skiing

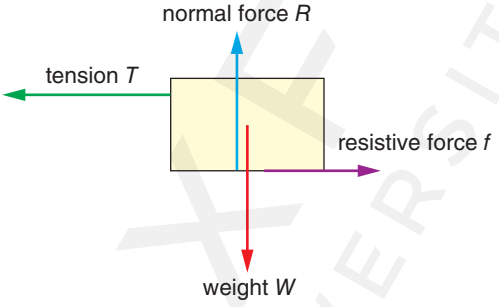
Susan of mass 55 kg is pulled by a horizontal rope and accelerates towards the left along a straight path. The tension in the rope is 210 N. The resistive force acting on her by water and air is 100 N. Her velocity at time $t = 0$ is u.

(a) What is Susan's acceleration?

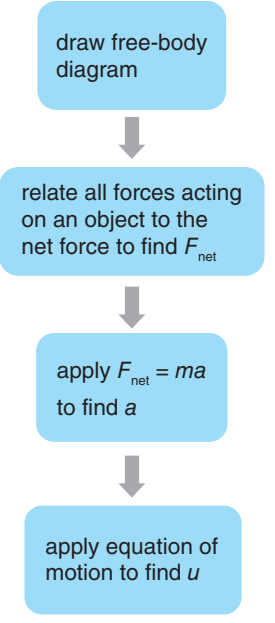
(b) Susan travels 6 m from $t = 0$ to $t = 1$ s. Find u.

(c) Susan lets go of the rope suddenly. Find her acceleration right after letting go.

Solution



Flow chart for parts (a) and (b):



Take leftwards as positive.

(a) $F_{net} = T + (-f) = 210 + (-100) = 110$ N

By $F_{net} = ma$,

$$a = \frac{F_{net}}{m} = \frac{110}{55} = 2 \text{ m s}^{-2}$$

Susan's acceleration is 2 m s^{-2} (leftwards).

(b) By $s = ut + \frac{1}{2}at^2$,

$$6 = u \times 1 + \frac{1}{2} \times 2 \times 1^2$$

$$u = 5 \text{ m s}^{-1}$$

u is 5 m s^{-1} (leftwards).

(c) $F_{net} = -f = -100$ N

$$a = \frac{F_{net}}{m} = \frac{-100}{55} = -1.82 \text{ m s}^{-2}$$

Susan's acceleration is 1.82 m s^{-2} (rightwards).

Tip

F_{net} acts along Susan's moving direction. We need to consider only the force(s) acting along this direction when applying $F_{net} = ma$. Note that T and f represent the magnitudes of forces. You need to add a negative sign if a force points to the negative direction.

Tip

After Susan lets go of the rope, the tension no longer acts on Susan and the resistive force becomes the net force acting on Susan.

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 William pushes a toy car and the toy car is moving to the left (Fig a). The total mass of the toy car and the boy sitting in it is m.

Fig a

Take leftwards as positive. Complete the table.

	F / N	f / N	m / kg	a / m s^{-2}
(a)	50	0	40	
(b)		40	30	−0.5
(c)	40		35	1

Learn by practice 2

1 Figure a shows the free-body diagram of a 2-kg object. The acceleration of the object is 7 m s^{-2} (upward).

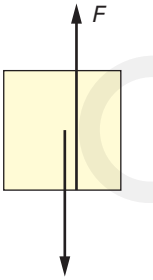


Fig a

(a) Find the net force acting on the object.

(b) Find the magnitude of force F.

2 A car is pulled by a tow truck on a straight road with an initial velocity of 10 m s^{-1}. The mass of the car is 1000 kg. Figure b shows the free-body diagram of the car.

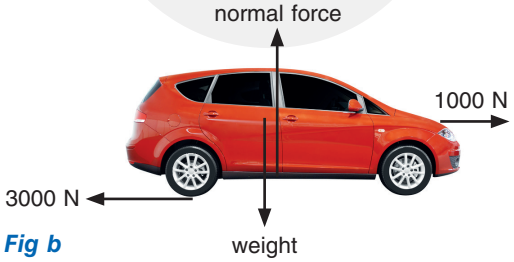


Fig b

(a) Find the net force acting on the car.

(b) Find the acceleration of the car.

(c) Find the final velocity of the car after it has been moving for 20 m.

3 A woman and her daughter push a 1200-kg car forwards along a rough horizontal surface. The woman pushes with 400 N and her daughter pushes with 50 N. The car is initially at rest. After being pushed for 10 s, the car has moved 5 m.

Fig c

(a) Draw the free-body diagram for the car.

(b) Find the acceleration of the car.

(c) Find the net force acting on the car.

(d) Determine the magnitude of the friction acting on the car.

Exercise 3.3

1 A student plots a graph of the net force F acting on an object against the acceleration a of the object (Fig a). The mass of the object is m.

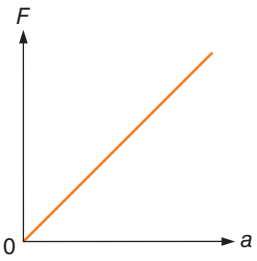


Fig a

What is the physical meaning of the slope of the graph?

A $\frac{1}{m^2}$ B $\frac{1}{m}$

C m D m^2

2 A woman pushes a wheelchair with a force of 50 N along a horizontal road and the wheelchair speeds up at 0.5 m s^{-2} (Fig b). The mass of the wheelchair is 70 kg.

Fig b

What is the net force acting on the wheelchair?

A 25 N B 30 N

C 35 N D 40 N

***3** A force F acts on an object of mass m. The acceleration of the object is a. Which of the following statements is/are correct?

(1) a is always directly proportional to F.

(2) a and F always point in the same direction.

(3) a may be zero when F is not zero.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

***4** Explain whether each of the following statements is correct.

(a) An object moves at a constant velocity when it is acted on by a constant net force.

(b) An object always moves in the direction of the net force acting on it.

(c) For an object being thrown upwards, the net force acting on it is zero when it reaches its highest position.

***5** A girl pushes a box of mass 60 kg forwards on the horizontal floor with a constant horizontal force (Fig c). The box speeds up uniformly from rest to 1 m s^{-1} in 2 s. The friction acting on the box is 50 N.

Fig c

(a) Find the pushing force by the girl.

(b) Find the acceleration of the box if the pushing force by the girl becomes 120 N.

***6** A block of mass 0.8 kg moves at an initial velocity of 5 m s^{-1} along a straight line on a rough horizontal plane (Fig d). It stops after moving 10 m.

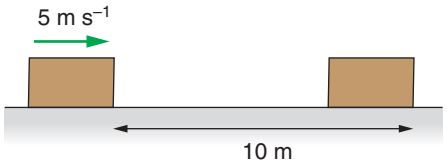


Fig d

(a) Draw the free-body diagram for the block when

(i) it is moving,

(ii) it is at rest.

(b) Find the acceleration of the block when it is moving.

(c) Find the friction acting on the block when it is moving.

3.4 Weight and Newton's second law of motion

DSE goal

Distinguish between mass and weight.

Realize the relationship between mass and weight.

DSE exam

14(1B)Q4(a)–(b), 16(1B)Q3, 17(1A)Q7, 23(1A)Q8

In daily life, we always consider kilogram as the unit of weight (Fig 3.4a).

Fig 3.4a *Kg is not a unit of weight in science.*

However, it is wrong in science. In fact, the kilogram is a unit of mass while the newton is a unit of weight!

1 Weight

When an apple of mass m falls freely near the Earth's surface, it moves with the acceleration due to gravity g. It is only acted on by its weight W, which is the gravitational force acting on it by the Earth.

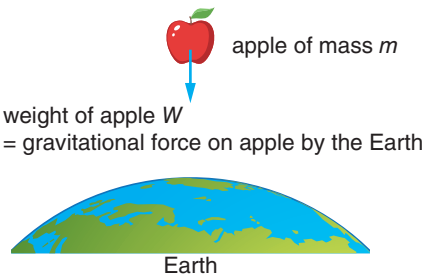


Fig 3.4b. *Weight is the gravitational force acting on an object.*

By $F_{net} = ma$, $W = mg$.

The above relationship always holds, no matter whether an object is falling or not. **Near the Earth's surface, g = 9.81 m s^{-2}**. Therefore, the weight of a 1-kg object is

$$W = mg = 1 \times 9.81 = 9.81 \text{ N} \approx 10 \text{ N}$$

Noted that near the Earth's surface, weight always points downwards.

The weight of an object on the Earth is the gravitational force acting on it by the Earth.

weight (N) **mass** (kg) **acceleration** due to gravity (m s^{-2})

$$W = mg$$

2 Mass and weight

Mass is a measure of an object's inertia. The mass of an object depends on the amount of matter in the object. It is the same no matter where you take the measurement.

By contrast, the weight of an object depends on its location because the gravitational force can vary greatly at different places, such as on different planets (Fig 3.4c).

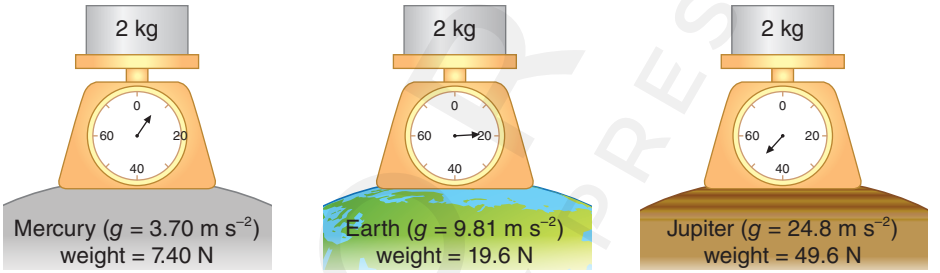


Fig 3.4c *The weight of the same mass varies on different planets.*

	Mass m	Weight W (= mg)
Meaning	**measure of inertia of object**	**gravitational force acting on object**
Unit	**kilogram (kg)**	**newton (N)**
Scalar or vector?	**scalar**	**vector**
Does it change?	**no**	**yes, depends on location**

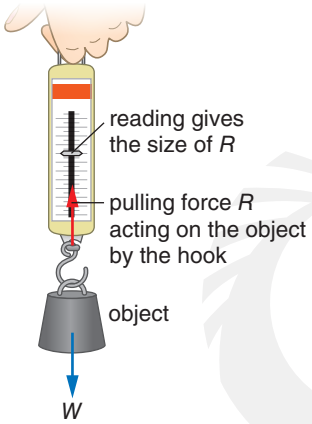


Fig 3.4d *Using a spring balance to measure weight.*

Weight is a force, and hence can be measured by a spring balance. The reading shows the size of the pulling force acting on the object by the hook, and should be equal to the weight of the object when held stationary (Fig 3.4d). If the measurement is taken near the Earth's surface, the mass of the object can then be derived by $m = \frac{W}{g}$ where g = 9.81 m s^{-2}.

Besides the spring balance, there are many types of weighing scales in daily life (Fig 3.4e). Although they are calibrated in kg, they actually measure weight.

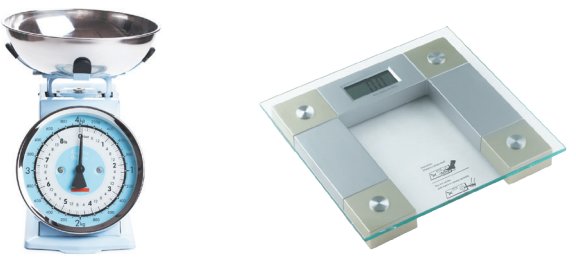
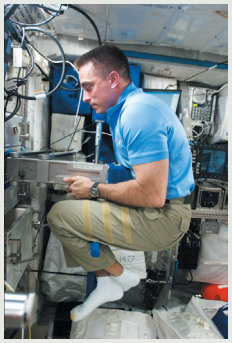


Fig 3.4e *Different weighing scales.*

Know more

Measuring the mass of an astronaut

Astronauts in space measure their masses by using the device below. The device pulls an astronaut with a constant force and the acceleration of the astronaut is measured. The mass is then calculated by $F = ma$.



Example 5 Weight of an astronaut on the Moon

An astronaut has a mass of 80 kg. When he is on the Moon, his weight is one-sixth of that on the Earth.

(a) Find his weight on the Earth and on the Moon.

(b) Find the acceleration due to gravity g_M on the Moon.

Solution

(a) Weight on the Earth $W_E = mg = 80 \times 9.81 = 785$ N

Weight on the Moon $W_M = \frac{1}{6} \times W_E = \frac{1}{6} \times 785 = 131$ N

(b) $g_M = \frac{W_M}{m} = \frac{131}{80} = 1.64$ m s^{-2}

Example 6 Worker on a platform

A 60-kg worker is standing on a platform (Fig a). Figure b shows the free-body diagram for the worker.

Fig a

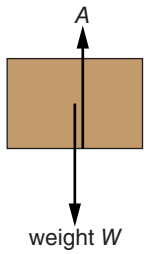


Fig b

(a) Find the magnitude of W and name force A.

(b) Find the magnitude of A when the platform is moving upwards

(i) with a constant velocity,

(ii) with an acceleration of 0.2 m s^{-2}.

Solution

(a) $W = mg = 60 \times 9.81 = 589$ N

A is the normal force exerted by the platform.

(b) (i) $A = W = 589$ N

(ii) Take upwards as positive. By $F_{net} = ma$,

$A + (-W) = ma$

$A - 589 = 60 \times 1.2$

$A = 661$ N

Tip

No net force during uniform motion ▶

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 Complete the following statements about mass and weight.

(a) Mass is measured in (kg / N).

(b) Weight is measured in (kg / N).

(c) The mass of an object (changes / does not change) with its position.

(d) The weight of an object (changes / does not change) with its position.

2 An object of mass m is on a planet where the acceleration due to gravity is g. Its weight is W. Complete the table below.

	m / kg	g / m s^{-2}	W / N
(a)	65	9.81	
(b)		3.72	5.58
(c)	4		44.6
(d)	2	0.62	

Learn by practice 3

1 At lift-off from the Earth's surface, the Falcon-9 rocket has an initial acceleration of 3.49 m s^{-2} upwards. The mass of the rocket is 5.72×10^5 kg.

Fig a

There are two forces acting on the rocket: weight and thrust (the upward force).

(a) Draw the free-body diagram for the rocket.

(b) Find the weight of the rocket.

(c) Find the thrust.

2 In everyday language, people say 'a bag of sugar weighs 1 kg'. This statement is improper in physics. Rewrite the statement in a proper way.

3 A spacecraft is launched upwards from the surface of the Earth into the sky. The mass of the spacecraft is 8000 kg. The upward thrust exerted on the spacecraft is 200 000 N.

(a) Find the gravitational force acting on the spacecraft by the Earth.

(b) Find the acceleration of the spacecraft.

4 A 0.2 kg figure is hung from a string (Fig b). At a certain moment, the figure is moving downwards and the tension acting on it is 2.1 N pointing upwards. Find the acceleration of the figure.

Fig b

3 Feeling of weight

A video showing the change in the weight of a girl measured by a weighing scale in a lift:

Video

Have you ever felt a loss or gain in weight when travelling in a lift? Does your weight really change?

Consider a girl standing on a weighing scale. The scale exerts an upward normal force R on her (Fig 3.4f). The scale reading gives the size of R, instead of the weight W acting on the girl.

If the scale is calibrated in kg, the reading should be divided by g to obtain the size of R.

→ Phone experiment worksheet for 'Acceleration of a lift' is available on OUP web.

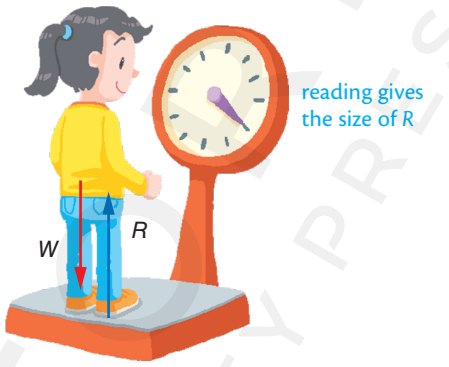


Fig 3.4f *Measuring weight using a weighing scale.*

R is not the same as *W* when the lift accelerates. Ts may ask Ss if this violates Newtons 3rd law after teaching Ch 3.7.

When the girl and the scale are **stationary**, the net force acting on the girl is zero and $\boldsymbol{R = W}$. In this case, the scale reading is the same as the girl's weight (Fig 3.4g).

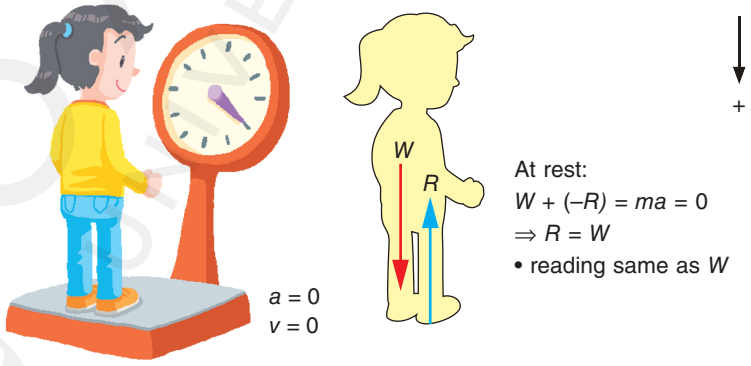


Fig 3.4g *A girl on a stationary scale and her free-body diagram.*

See Example 6 on p.119.

However, if the girl measures her weight in an accelerating lift, R will not be equal to W since the net force acting on the girl is no longer zero.

Suppose the lift moves from the top floor to the ground floor. Since g is constant near the Earth's surface, her weight W (= mg) is also constant. However, the scale reading (= R) changes.

The girl's feeling of her own weight depends on R. In a lift travelling with an acceleration, she does not feels her normal weight W, but her apparent weight (whose size is given by R). Figure 3.4h on p.122 further illustrates the above discussion.

apparent weight 表觀重量

Are these still correct if the girl is in a lift that goes up?

Simulation

→ This simulation explains the cause for apparent weight in a lift. It shows how the change in motion of the lift (at rest, constant speed, acceleration or deceleration) affects the normal force on the person by the lift.

▶ We can see that:

1 She feels a loss in weight when her acceleration points downwards.

2 She feels the normal weight when she moves uniformly.

3 She feels a gain in weight when her acceleration points upwards.

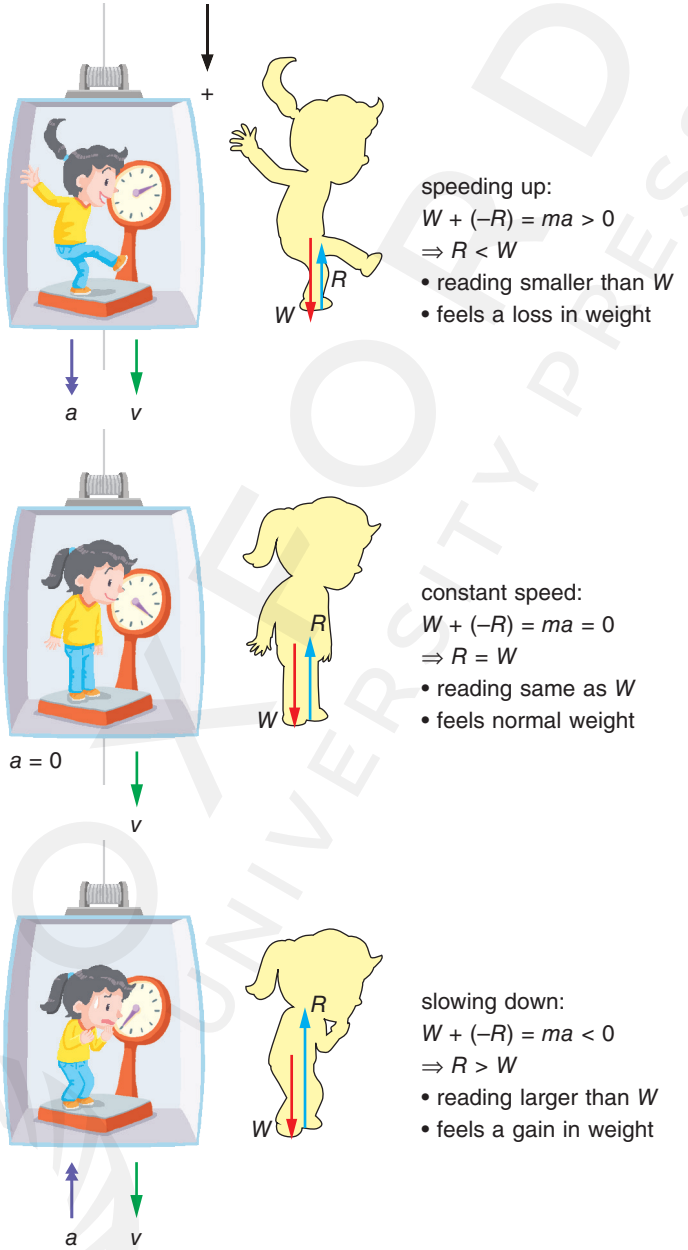


Fig 3.4h *Measuring weight in a lift that goes down (downwards as positive).*

In the extreme case when the lift falls freely, the girl falls with an acceleration g.

$$W + (-R) = ma$$

$$mg - R = mg$$

$$R = 0$$

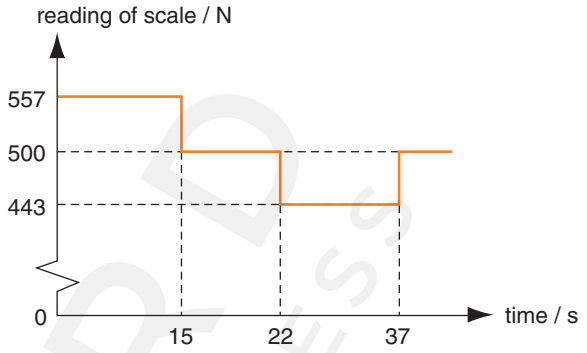
A weighing scale will show zero reading in this case.

▶ In this case, the girl **feels** *weightless*.

weightless 失重

Example 7 Feeling of weight in a lift

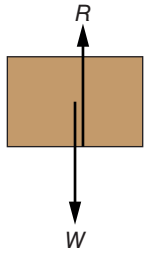
In a stationary lift at a high tower, a girl steps on a weighing scale calibrated in newtons. The scale reads 500 N. The lift then starts from rest at $t = 0$ and takes 37 s to travel from the 5th floor to the 89th floor. The reading of the scale is as shown. Find the acceleration of the lift from (a) 0–15 s, and (b) 22–37 s.



Solution

Take upwards as positive.

$$m = \frac{W}{g} = \frac{500}{9.81} = 50.97 \text{ kg}$$



(a) From 0–15 s,

$F_{net} = R + (-W) = ma$

$557 + (-500) = 50.97a$

$a = 1.12 \text{ m s}^{-2}$

The acceleration of the lift from 0–15 s is 1.12 m s^{-2} (upwards).

◀ The result shows that the girl in a lift that goes up feels a gain in weight when her acceleration points upwards.

(b) From 22–37 s,

$F_{net} = R + (-W) = ma$

$443 + (-500) = 50.97a$

$a = -1.12 \text{ m s}^{-2}$

The acceleration of the lift from 22–37 s is 1.12 m s^{-2} (downwards).

◀ The result shows that the girl in a lift that goes up feels a loss in weight when her acceleration points downwards.

Quick check 7

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 A girl holds a spring balance in a lift (Fig a). An object of mass m is hung from the balance and the acceleration of the object is a. Take upwards as positive. Complete the following table by applying $R - W = ma$.

		R / N	W / N	m / kg	a / m s^{-2}
(a)	The lift is stationary.			1.2	
(b)	The lift is travelling downwards and is speeding up at 1.5 m s^{-2}.			1.2	
(c)	The lift is travelling downwards and is slowing down at 0.5 m s^{-2}.			1.2	

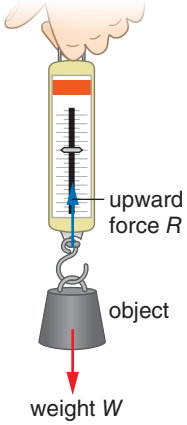


Fig a

Exercise 3.4

★**1** An old lift is packed with people and is almost overweight (Fig a). At which of the following instants would the overweight alarm most probably sound?

Fig a

(1) The lift slows down in the upward journey.

(2) The lift speeds up in the downward journey.

(3) The lift slows down in the downward journey.

A (1) only

B (3) only

C (1) and (2) only

D (1), (2) and (3)

★**2** A student is standing in a lift. The lift is moving downwards and speeding up at 3 m s^{-2}. The mass of the student is 65 kg.

(a) Draw the free-body diagram for the student.

(b) Find the net force acting on the student.

(c) Find the apparent weight of the student at this moment.

★**3** A 10-kg block is placed in a lift. The lift is moving upwards and speeding up.

(a) Explain why the apparent weight of the block is larger than the its weight in this case.

(b) The apparent weight of the block is 121.1 N. Find the acceleration of the lift.

★**4** A 5000-kg spacecraft is landing on the Moon. The acceleration due to gravity on the Moon's surface is 1.6 m s^{-2}. To land safely, a constant upward thrust is produced so that the spacecraft can slow down. The initial velocity of the spacecraft is 15 m s^{-1} downwards. The spacecraft has moved 20 m downwards in 10 s.

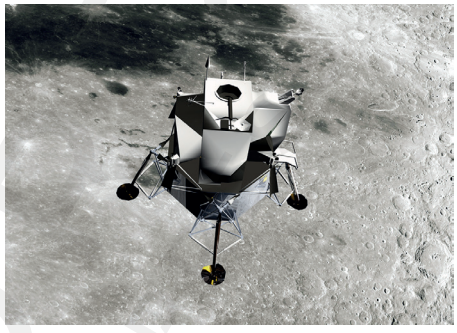


Fig b

(a) Draw the free-body diagram for the spacecraft.

(b) Find the acceleration of the spacecraft.

(c) Estimate the upward thrust.

★**5** An object of mass 2 kg is hung stationary from a spring balance (Fig c).

Fig c

(a) What is the reading taken on the Earth's surface?

(b) What is the reading taken on Venus's surface? The acceleration due to gravity on Venus's surface is 8.87 m s^{-2}.

(c) If the spring balance is released so that both the spring balance and the object fall with the acceleration due to gravity, what is the reading of the spring balance?

3.5 Fluid resistance

A flying squirrel can land safely even if it falls from a great height. How does it work? What is the physics behind it?

DSE exam
17(1B)Q4(b)

Video

→ This video shows a flying squirrel falling from a great height.

Fig 3.5a *A flying squirrel.*

The answer is related to a force called *fluid* resistance. Objects experience resistive forces when they move through fluids, i.e. gases or liquids. Air resistance is an example of fluid resistance.

Fluid resistance increases with the speed of the moving object and is **zero if the object is at rest** in the fluid. *Streamlined* bodies can reduce the fluid resistance (Fig 3.5b).

Smoke (the white lines) is added to visualize the air flow around a car in a wind tunnel. ▶

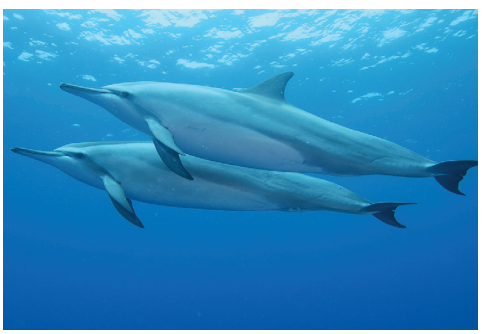


Fig 3.5b *Streamlined bodies.*

fluid 流體　　streamlined 流線型

A video showing skydiving:

Video

Know more

Skydiving terminal speed

Normally, the terminal speed of a skydiver is about 50–90 m s^{-1}. *Felix Baumgartner* achieved 377 m s^{-1} when he dove from a height of 39 km. The atmosphere is less dense at that height, resulting in a smaller air resistance.

The following video shows the dive.

Video

The flying squirrel on p.125 stretches its parachute-like membrane to increase the air resistance during falling. It can fall at a low terminal speed and prevents itself from being injured in landing.

Wingsuit fliers resemble flying squirrels.

Video

→ This video shows some wingsuit fliers flying.

✎ Alan Eustace made a jump from a height of 41.4 km on 24 Oct 2014 and broke Baumgartner's world record for the highest-altitude jump. However, his terminal speed (367 m s^{-1}) was lower than Baumgartner's.

▶ Let us take skydiving as an example to study the effect of fluid resistance. Under the action of the body weight and air resistance, the motion of a skydiver can be divided into several stages (Table 3.5a).

① Suppose the skydiver falls from rest just after leaving the aeroplane. He experiences no air resistance and his weight W is the net force F_{net} acting on him. At this moment, his acceleration a is equal to g.	① W
② As the skydiver gains speed, the air resistance f acting on him increases. F_{net} (= $W - f$) decreases and keeps pointing downwards, and so does a (by $F_{net} = ma$, $F_{net}\downarrow \Rightarrow a\downarrow$). Note that the skydiver is still speeding up, but at a lower rate.	② f W
③ f increases until it balances W. F_{net} becomes zero and so does a ($F_{net} = W - f = 0 \Rightarrow a = 0$). The skydiver falls with a constant speed called **terminal speed**.	③ f W
④ The skydiver keeps on falling at the terminal speed.	④ f W
⑤ After the parachute is opened, f increases sharply. F_{net} (= $W - f$) points upwards and so does a. The skydiver slows down to a speed (< 5 m s^{-1}) safe for landing. ✎ Remind Ss about the difference between a decreasing acceleration and a deceleration. The skydiver decelerates only after opening the parachute.	⑤ f W

Table 3.5a *Different stages in skydiving (downwards as positive).*

▶ The v–t and a–t graphs of the skydiver are shown in Figure 3.5c.

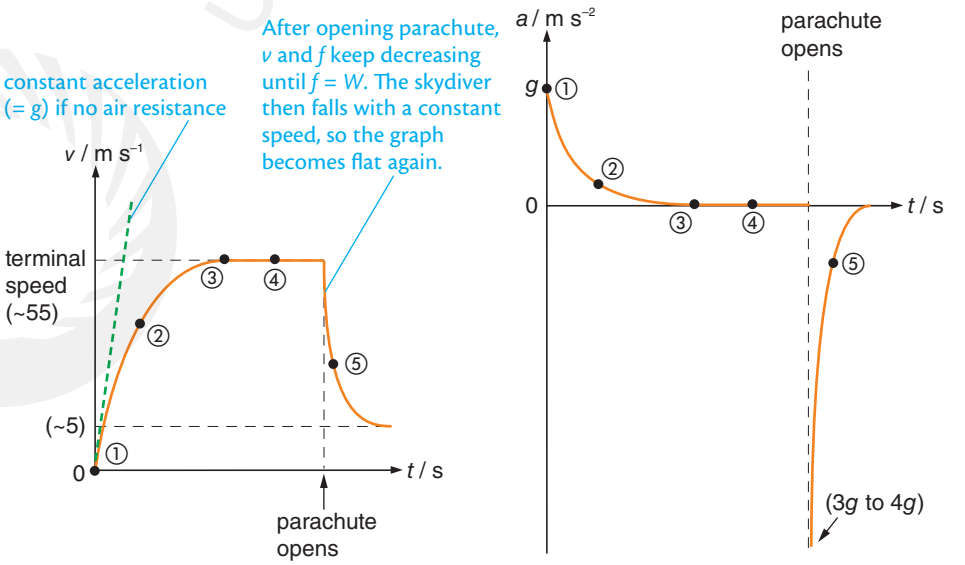


Fig 3.5c *The v–t and a–t graphs of a skydiver (downwards as positive). The deceleration can be as high as 40 m s^{-2} after opening the parachute.*

terminal speed 終端速率　　Felix Baumgartner 保加拿

Know more

Terminal speed of raindrop

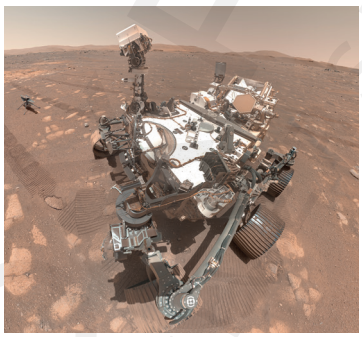
Raindrops have different terminal speeds. A small drop of diameter 0.5 mm has a terminal speed of about 2 m s^{-1} while a large drop of diameter 5.0 mm has a terminal speed of about 9 m s^{-1}.

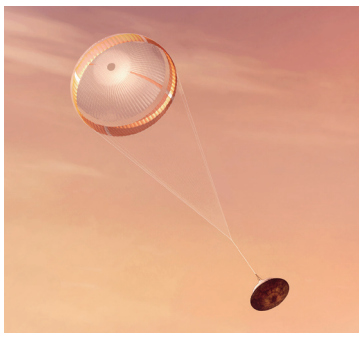
The motion of an object falling in a fluid is summarized as follows:

- **Under the effect of fluid resistance, an object falls with decreasing acceleration until the acceleration becomes zero, and the object falls with a maximum constant speed (terminal speed).**
- **The fluid resistance acting on an object increases with its speed. When the fluid resistance equals to its weight, the net force acting on it becomes zero. By Newton's first law of motion, it falls with a constant speed.**

Example 8 Landing of Perseverance

Perseverance is a robotic rover sent to Mars. The spacecraft holding Perseverance was slowed down by fluid resistance in the atmosphere of Mars. When it was 11 km above the surface of Mars, a parachute was opened.





(a) Suppose the spacecraft was moving vertically downwards with a deceleration of 50 m s^{-2} at a certain instant. Take the mass and weight of the spacecraft as 3890 kg and 14 400 N respectively. Find the fluid resistance acting on the spacecraft.

(b) How does the fluid resistance acting on the spacecraft change with time after the parachute is opened?

Solution

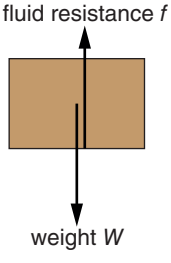
(a) Take upwards as positive.

By $F_{net} = ma$,

$$f + (-W) = ma$$

$$f + (-14\,400) = 3890 \times 50$$

$$f = 209\,000 \text{ N}$$



(b) The fluid resistance decreases until reaching a value equal to the weight of the spacecraft. Then it becomes constant.

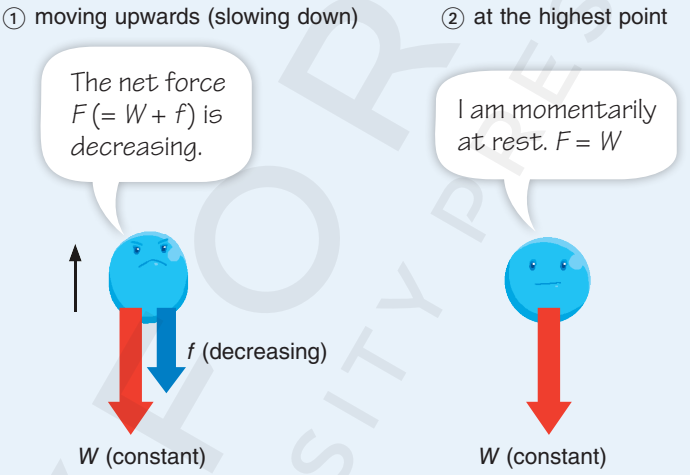
Perseverance 毅力號

Right and wrong

A ping-pong ball is thrown upwards and air resistance is **not** negligible. The net force acting on it during its upward journey

- ✗ remains constant.
- ✗ keeps decreasing until reaching zero.
- ✓ keeps decreasing and is equal to the ball's weight at the highest point.

Suppose the ball's weight is W and air resistance is f. The figure below explains how the net force changes.



Google Forms Microsoft Forms

Quick check 8

→ E-version in OUP Exercise Platform is also available.

1 Paper ball X and paper Y are dropped from the same height (Fig a). They have the same mass of 0.005 kg.

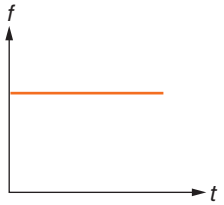
Fig a

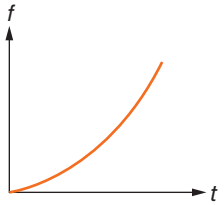
(a) Which paper reaches the floor first?

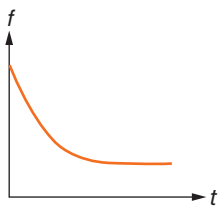
(b) Find the air resistance acting on X and Y after they have reached their terminal speeds.

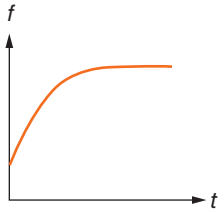
Exercise 3.5

1 A skydiver falls at the terminal speed before opening his parachute. Which of the following graphs best represents how the air resistance f acting on him varies with time t after he has opened his parachute?

A 

B 

C 

D 

***2** A ping-pong ball is thrown vertically upwards at time $t = 0$. It falls back to the original position at $t = T$.

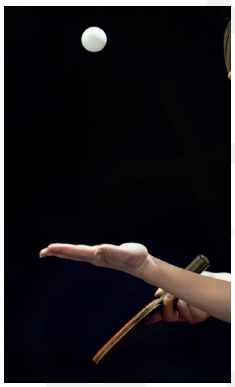
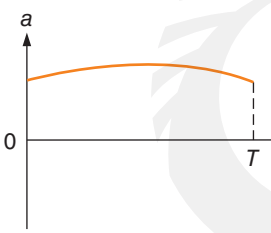
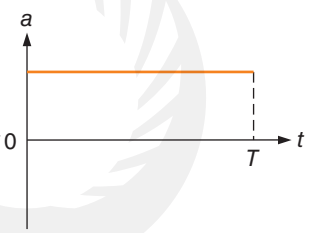
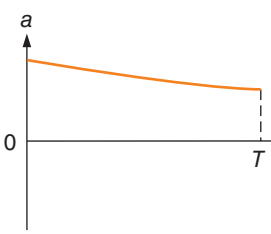


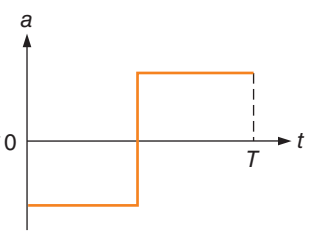
Fig a

Which of the following graphs best represents how the acceleration a of the ball varies with t? Take downwards as positive.

A 

B 

C 

D 

***3** A lemon falls from rest in water at time $t = 0$ (Fig b). It reaches the terminal speed at $t = T$.

Fig b

Which of the following statements is/are correct?

(1) The net force acting on the lemon points upwards from $t = 0$ to $t = T$.

(2) The weight of the lemon is zero at $t = T$.

(3) The lemon speeds up from $t = 0$ to $t = T$.

A (3) only B (1) and (2) only

C (1) and (3) only D (1), (2) and (3)

***4** When a raindrop falls, two forces A and B act on it (Fig c).

(a) Name the forces A and B.

(b) Explain why the raindrop falls with a constant speed after a certain time.

Fig c

***5** A skydiver opens his parachute after he reaches the terminal speed.

(a) Describe and explain his motion after he opens his parachute.

(b) The mass of the skydiver is 70 kg. The air resistance acting on him is 3200 N just after he opens his parachute. Find his deceleration at that moment.

Fig d

***6** A raindrop falls from rest.

(a) Gloria says, 'Since air resistance acts on the raindrop, it slows down as it falls.' Do you agree with her? Explain briefly.

(b) The raindrop reaches the terminal speed before hitting the ground.

(i) Sketch a graph to show how the net force F acting on the raindrop varies with time t during its fall.

(ii) When is the magnitude of acceleration of the raindrop the largest during its fall?

3.6 Friction

DSE exam
12(1A)Q11, 13(1A)Q7, 22(1A)Q5, 23(1B)Q3(a)

We learned that friction arises when two objects are sliding or tend to slide over one another. Unlike fluid resistance, **friction can act on both moving objects and stationary objects** (Fig 3.6a).

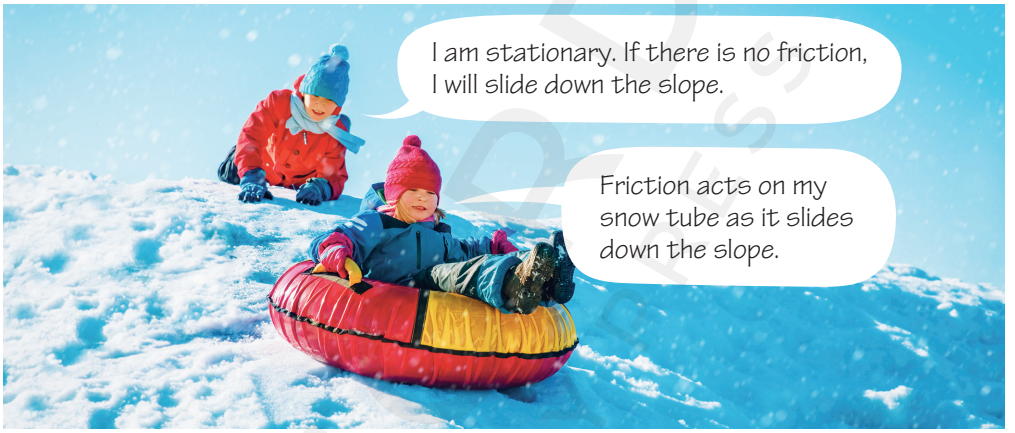


Fig 3.6a *Friction can act on both moving objects and stationary objects.*

Know more

More about friction

Friction is sometimes inconvenient, so people need to find ways to reduce it (Fig a).

(i) Using lubricant.

(ii) Using an air cushion.

Fig a

On the other hand, friction is essential to our daily life. For example, your foot pushes back when you walk. The ground exerts friction that points forwards on your foot. This friction propels you forwards (Fig b). Friction is also essential for a pencil to leave a mark on a piece of paper (Fig c).

Fig b

Fig c

The ground is a fixed surface. The case for a box on a movable surface will be discussed in Ch 4.

▶ Consider a box rests on a **rough horizontal ground**. Then a horizontal force F acts on it. F increases graudally from zero and moves the box finally. We will see how the magnitude of the **friction *f*** acting on the box by the ground changes.

Stage 1: The box **remains at rest**.

- Obviously, f is zero when $F = 0$ (Fig 3.6b).

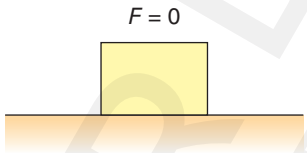


Fig 3.6b *f = 0 when F = 0.*

- When F is small, the box does not move. By Newton's first law of motion, the net force acting on the box is zero, i.e. f and F have the same magnitude but act in opposite directions.
- As F increases, ***f* increases accordingly**. This means that ***f* is not constant** (Fig 3.6c).

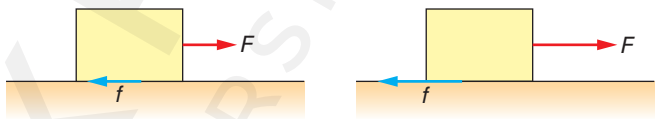


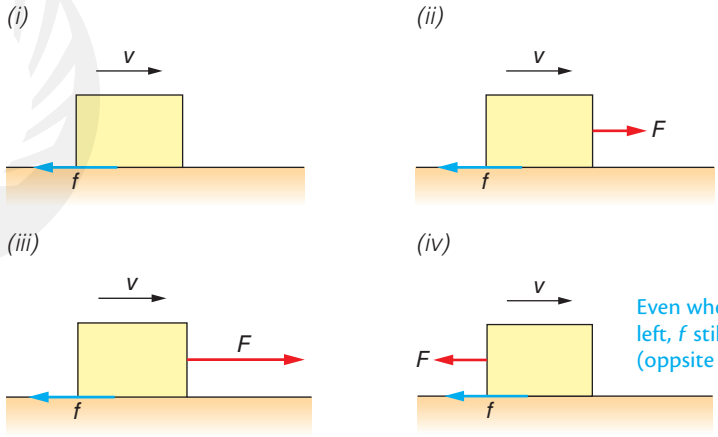
Fig 3.6c *When the box is at rest, f increases with F.*

- f does not keep increasing with F. It has a certain maximum value f_{max}. When $F > f_{max}$, the box starts to move.

Ss need not know that the static friction is slightly larger than the kinetic friction. However, they should know how the friction varies in different cases in order to answer questions like DSE 2012 Paper 1A Q11.

Stage 2: The box is **moving**.

- Once the box is moving, ***f* remains constant** and does not increase anymore. Its direction is **always opposite to the motion of the box**. Its magnitude and direction are independent of F (Fig 3.6d).



Even when F points to the left, f still points to the left (oppsite to the motion).

The graph of friction f against applied force F **when the box is moving**:

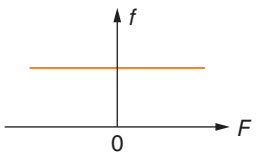
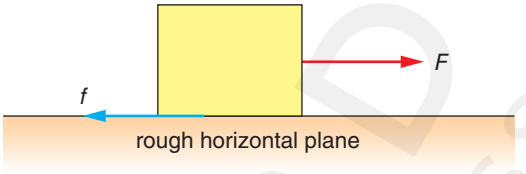


Fig 3.6d *When the box is moving, f is constant and always opposite to v.*

Example 9 Friction and applied force

A 2-kg box resting on a rough horizontal plane is pulled towards the right by a horizontal force F. The friction acting on it is f.



The box remains stationary when F is 1 N and accelerates at 3 m s^{-2} to the right when F = 10 N.

(a) Find the magnitude of f when F is 1 N.

(b) Find the magnitude of f when F is 10 N.

(c) Then F is reduced continuously from 10 N to 0 in a short period of time. The box keeps moving to the right within this period.

(i) Find the acceleration a of the box when F = 0.

(ii) Sketch a graph of a against F within the above period.

Solution

Take rightwards as positive.

(a) Since the box is stationary, by Newton's first law of motion,

$f = F = 1$ N

(b) By $F_{net} = ma$,

$F + (-f) = ma$

$10 - f = 2 \times 3$

$f = 4$ N

(c) (i) When F = 0, the box keeps moving to the right.

$\therefore f = 4$ N and points leftwards.

By $F_{net} = ma$,

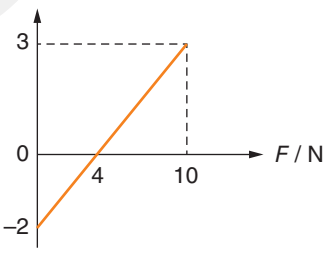
$F + (-f) = ma$

$0 + (-4) = 2a$

$a = -2$ m s^{-2}

The acceleration is 2 m s^{-2} (leftwards).

(ii) a / m s^{-2}



Tip

$F + (-f) = ma$

$F + (-4) = 2a$

$a = \frac{1}{2}F - 2$

a varies linearly with F.

When F = 0, a = –2 m s^{-2}

When F = 10 N, a = 3 m s^{-2}

Google Forms Microsoft Forms

Quick check 9

→ E-version in OUP Exercise Platform is also available.

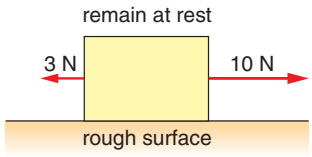
1 *True or false*:

(a) When an object is at rest, the friction acting on it must be zero. (T/F)

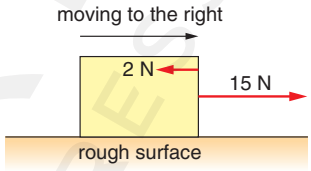
(b) Friction opposes the sliding motion of an object on a surface. (T/F)

2 Each of the following figures shows a block on a rough horizontal surface acted on by different horizontal forces. State the direction of friction acting on the block in each case.

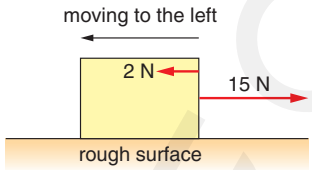
(a)



(b)



(c)



Exercise 3.6

★**1** A 2-kg box is pulled towards the right on a horizontal table by a horizontal force F.

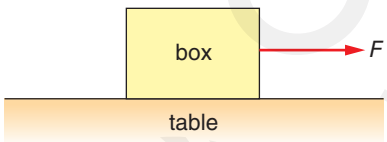


Fig a

(a) The box moves with a constant velocity when F is 3 N. What is the friction between the box and the table in this case?

(b) At a certain moment, the box is moving to the right. Find the acceleration of the box if

(i) F becomes 2 N,

(ii) F is removed.

★**2** A block is initially moving to the right on a rough horizontal surface and the friction acting on it is 5 N. A horizontal force of 12 N towards the left is suddenly applied onto the block and the block slows down at 2 m s^{-2}. Find the mass of the block.

★**3** A 3-kg box resting on a rough horizontal table is pulled towards the right by a horizontal force F, which increases gradually from zero. The friction acting on the box is f. Figure b shows the relationship between the magnitudes of f and F.

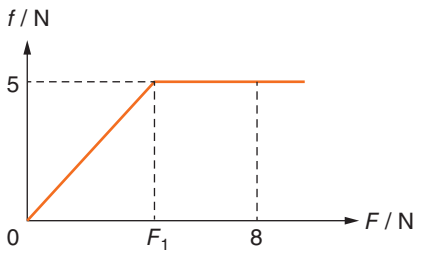


Fig b

(a) Find F_1.

(b) Find the acceleration of the box when F = 8 N.

(c) Describe the motion of the box when F increases from 0 to 8 N.

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3.7 Newton's third law of motion

1 Newton's third law of motion

DSE goal

Realize forces acting in pairs.

State Newton's third law of motion.

Identify action-and-reaction pair.

DSE exam

15(1A)Q11, 16(1A)Q6, 21(1A)Q4, 22(1B)Q3(a), 23(1B)Q2(c)(i)

If you sit on a chair on wheels and push horizontally against an unmovable desk, you will move backwards (Fig 3.7a). It is just as if you are pushed backwards by someone else (Fig 3.7b). This suggests that when you push the desk, the desk pushes back. This is a very important property of force: **forces always occur in pairs.**

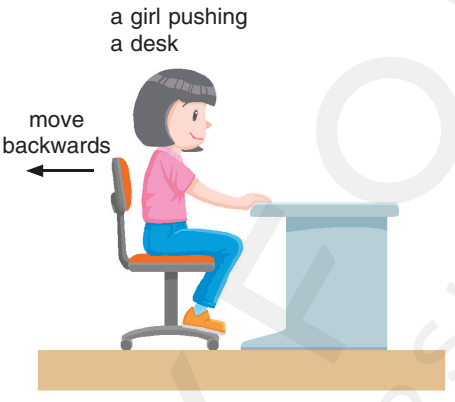


Fig 3.7a *You move backwards when pushing a desk.*

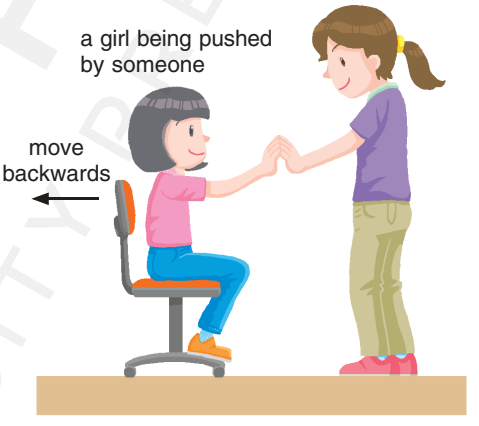


Fig 3.7b *You move backwards when being pushed by another person.*

Let us do the following experiment to study the paired forces in detail.

Experiment 3c / Paired forces

→ This video demonstrates Expt 3c. Video

1 Set up the apparatus as shown, with a rubber bumper attached to the force sensor of each trolley.

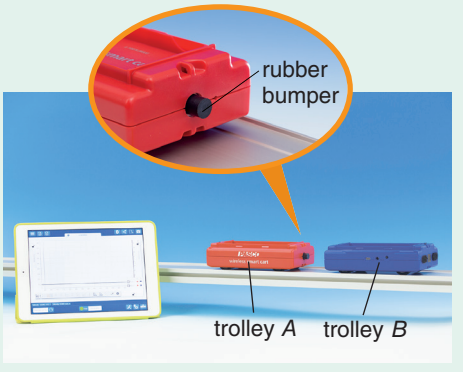
2 Take the direction to the right as positive when taking measurements. Start data-logging. Do the following:

(a) Push trolley *A* towards trolley *B*.

(b) Push both trolleys *A* and *B* towards each other.

In each case, note the force–time graphs (*F*–*t* graphs) obtained.

3 Repeat step 2 with (a) a harder push and (b) the mass of trolley *A* doubled.



Discussion

What do the *F*–*t* graphs show about the forces acting on the trolleys?

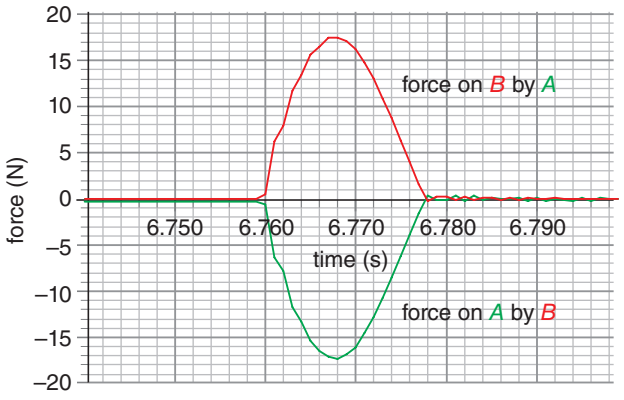


Fig 3.7c *F–t graph obtained in Experiment 3c.*

Figure 3.7c shows one of the *F–t* graphs obtained in Experiment 3c.

We can see that the force acting on *B* by *A* and the force acting on *A* by *B* appear **at the same time**. Also, the two forces have **equal magnitude** but are **opposite in direction** all the time. Other *F–t* graphs obtained in the experiment have the same features. This result demonstrates **Newton's third law of motion**:

For every *action*, there is an equal and opposite *reaction*. The action and reaction forces act on different interacting objects simultaneously.

Note the following about Newton's third law:

1 Forces always occur in pairs between **two** interacting objects.

2 The forces in a pair appear at the same time. Either force can be called the 'action' and the other one is then the 'reaction'. Naming a force 'action' does not imply that it causes the 'reaction'.

Example 10 | Launch of a rocket

Explain how a rocket is launched by ejecting hot gas downwards.

Solution

The rocket exerts a force on the gas to push the gas downwards. By Newton's third law, the gas exerts an equal but opposite reaction force to push the rocket upwards.

Ss may have the misconception that a rocket is propelled by the impact of exhaust gases against the ground. Ts may explain to Ss that a rocket can accelerate forwards in space when it ejects gases backwards.

Newton's third law of motion 牛頓運動第三定律　　action 作用力　　reaction 反作用力

Know more

Air drone and flying jet suit

To be able to fly, both an *air drone* (Fig a) and a *flying jet suit* (Fig b) rely on the force acted by the air. When they operate, they exert downward forces on the air. By Newton's third law, the air exerts equal and opposite forces on the machines to push them upwards (Fig c).

Fig a

→ This video shows the development of a kind of flying jet suit.

Video

Fig b

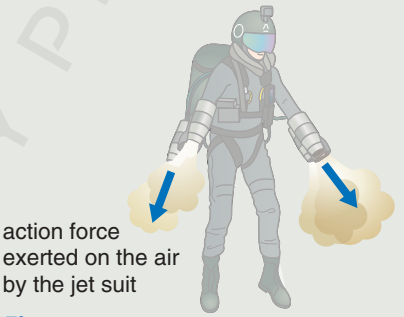


Fig c

Example 11 | Pushing a car

A man is pushing a car but the car does not move.

(a) If the pushing force on the car by the man is the action force, what is the reaction force?

(b) Explain why the car is stationary.

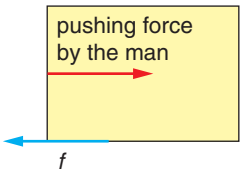
Solution

(a) The force acting on the man by the car

(b) The friction acting on the car by the ground and the pushing force by the man cancel each other out and therefore the net force acting on the car is zero. By Newton's first law, the car remains stationary.

Tip

Consider the forces acting on the car when analysing the motion of the car.



Think about...

Samuel says that the car is stationary because the pushing force on the car by the man is cancelled out by the reaction force in (a). Do you agree?

air drone 無人機　　flying jet suit 飛行套裝

Right and wrong

✗ When the man pulls the cart, the cart exerts an equal but opposite force that pulls the man back. Therefore, the cart does not move.

✓ Action and reaction do not act on the same body. They will not cancel each other out. The cart does not move because the pulling force by the man is cancelled out by the friction acting on the cart.

Google Forms Microsoft Forms

Quick check 10

→ E-version in OUP Exercise Platform is also available.

1 A bus collides with a car. Which vehicle exerts a greater force on the other?

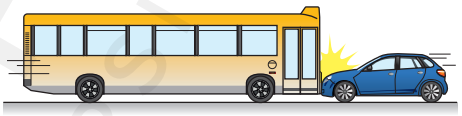


Fig a

A The bus

B The car

C The forces are equal in magnitude.

2 Rocket *X* ejects gas backwards onto the ground while rocket *Y* ejects gas backwards in space. Which rocket is acted on by a forward force?

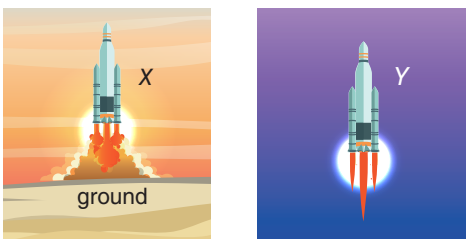


Fig b

A *X*

B *Y*

C Both of them

3 A bowling ball knocks down a pin (Fig c). Which object exerts a greater force on the other?

A The ball

B The pin

C The forces are equal in magnitude.

Fig c

2 Action-and-reaction pairs

We now know that an **action-and-reaction pair** should

1 act on two different objects that interact with each other,

2 have an equal magnitude at all times, and

3 act in opposite directions.

Ts may ask Ss to show that the force acting on the chair by the dog has the same magnitude as the weight of the dog.

Consider a dog sitting on a chair. The forces acting on it are its weight and the normal force by the chair (Fig 3.7d). Although these two forces are equal in magnitude and opposite in direction, they do not form an action-and-reaction pair because they act on the same object (the dog).

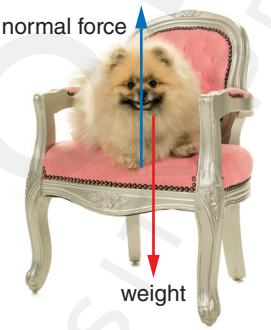
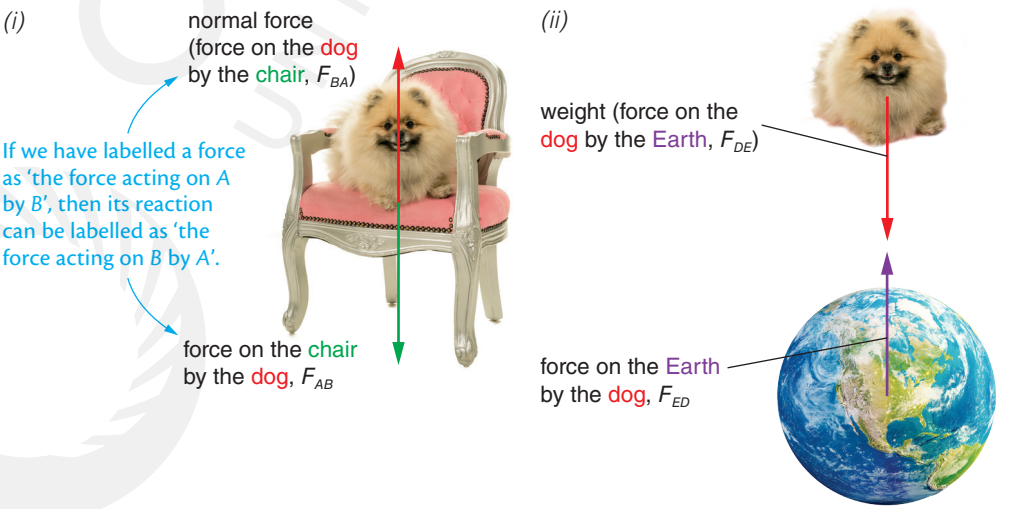


Fig 3.7d *Forces acting on a dog sitting on a chair.*

Figure 3.7e shows two action-and-reaction pairs involved in this situation. One pair is between the dog and the chair; the other is between the dog and the Earth.



Caution

The diagrams in Figure 3.7e are not free-body diagrams. A free-body diagram shows forces acting on **one** body only.

Fig 3.7e *Action-and-reaction pairs.*

From the above example, we can see that we cannot judge whether two forces are action-and-reaction pairs by looking at their magnitudes and directions only. Instead, we have to look at whether the two forces are the interaction between two objects.

action-and-reaction pair 作用力—反作用力對

Quick check 11

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Determine the action-and-reaction pairs among the given forces in the following cases.

(a) An owl resting on the ground

W: weight of the owl

R: normal force acting on the owl by the ground

F: gravitational force acting on the Earth by the owl

(b) A Christmas bauble hung from a string

W: weight of the bauble

T: tension exerted by the string on the bauble

F: force exerted by the bauble on the string

☆ HKDSE 2016 Paper 1A Q6

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Basket on the ground

A basket of weight *W* is at rest on the ground. Some helium balloons are tied to the basket and exert an upward force *F* on it. The normal force exerted on the basket by the ground is *R*. What is the magnitude of the force exerted on the ground by the basket?

A $W - R$

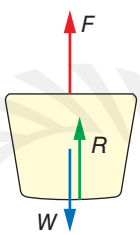
B $W + R$

C $W - F$

D $W + F$

Solution

The free-body diagram for the basket is as shown.



Common mistake

Students may have difficulty in identifying the forces acting on the basket.

Since the basket remains stationary, by Newton's first law,

$W = R + F$

$R = W - F$

The force exerted on the ground by the basket and *R* are an action-and-reaction pair. They have the same magnitude, i.e. $W - F$.

Common mistake

Students may have difficulty in identifying the action-and-reaction pair.

∴ The answer is C.

▶ Revision 3 Q21 (p.147)

Exercise 3.7

1 In an action-and-reaction pair, what would happen to the reaction if the action disappears?

A It continues acting on the object.

B It disappears at a later time.

C It disappears immediately.

D Its magnitude decreases gradually to zero.

***2** A woman pushes an iron with a force F on a horizontal ironing board (Fig a).

Fig a

(a) Figures b and c show the iron and the ironing board separately.

Fig b

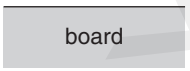


Fig c

Draw and label the following forces in the correct figures.

F: pushing force

W: weight of the iron

R: normal force acting on the iron by the board

f_1: friction acting on the iron by the board

f_2: friction acting on the board by the iron

(b) Which of the above forces form(s) action-and-reaction pair(s)?

***3** Describe how a cat jumps upwards using Newton's third law of motion (Fig d).

Fig d

***4** Wooden block X is placed on another wooden block Y, which is lying on the ground (Fig e). Figures f and g show their free-body diagrams.

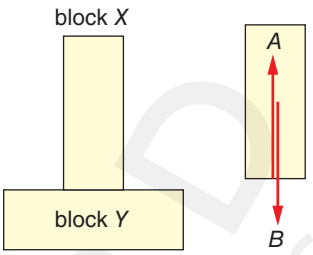


Fig e

Fig f

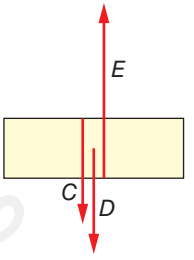


Fig g

(a) Complete the following descriptions for forces A to E.

A: Force acting on _____ by _____

B: Force acting on _____ by _____

C: Force acting on _____ by _____

D: Force acting on _____ by _____

E: Force acting on _____ by _____

(b) Which of the above forces form action-and-reaction pair(s)?

***5** Why can't you leave the ground no matter how hard you pull your trousers upwards (Fig h)? Explain briefly.

Fig h

***6** Objects X and Y are connected by string. X is held by hand as shown (Fig i).

(a) Draw the free-body diagrams for X and Y separately.

(b) Which forces in (a) form action-and-reaction pair(s)?

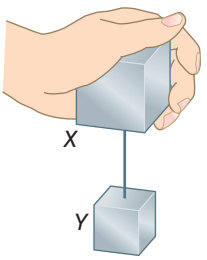


Fig i

***7** Suppose you are on a sailboat and you have a magic fan that can create a strong wind. Explain how you can propel the boat with the fan.

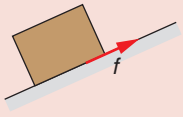
Key concepts 3

1 Force

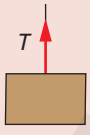
1 Unit: newton (N)
2 Vector quantity

2 Common types of force

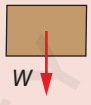
Friction (f)



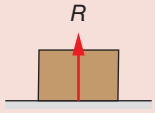
Tension (T)



Weight (W)

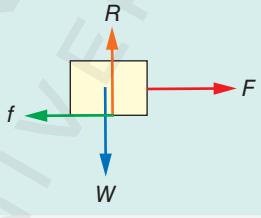


Normal force (R)



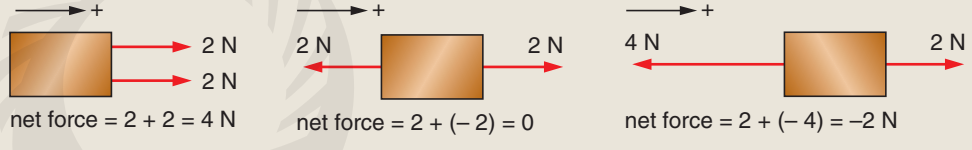
3 Free-body diagram

Showing all the forces acting on an object



4 Net force

Addition of parallel forces



5 Inertia and Newton's first law of motion

Inertia	Mass	Newton's first law of motion
The tendency of an object to maintain its state of rest or of constant velocity	Unit: kg Mass ↑ ⇒ Inertia ↑	Every object remains in a state of rest or uniform motion unless it is acted on by a net force.

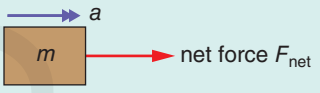
6 Newton's second law of motion

The acceleration a of an object is

1 directly proportional to, and in the same direction as, the net force F_{net} acting on it, and

2 inversely proportional to the mass m of the object.

$$F_{net} = ma$$

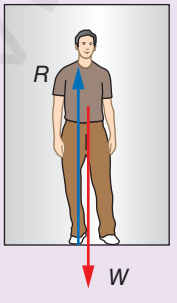


7 Mass and weight

Mass m	Weight W
measure of inertia of object	gravitational force acting on object, $W = mg$
unit: kg	unit: N
value does not depend on the location	value depends on the location
measured by beam balances	measured by spring balances / weighing scales

8 Feeling of weight

- Weight W is not affected by state of motion.
- The normal force R gives us the feeling of our own weight.

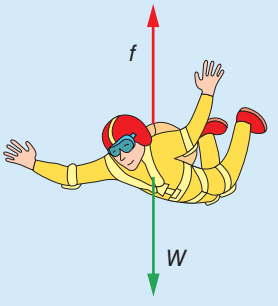


accelerates upwards
$R > W$
feels a gain in weight

accelerates downwards
$R < W$
feels a loss in weight

9 Fluid resistance

- The faster an object, the greater the fluid resistance f.
- An object falls with the terminal speed when $W = f$.

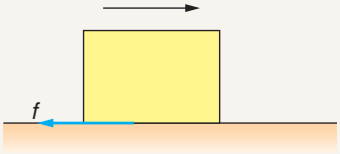


10 Friction

For an object resting on a rough horizontal surface:

External force F	Friction f
$F = 0$	$f = 0$
$F < f_{max}$	$f = F$
$F > f_{max}$	$f = f_{max}$

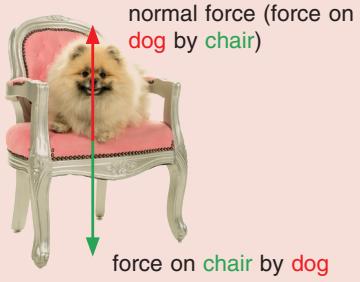
For an object moving **to the right** on a rough horizontal surface:



External force F	Friction f
rightwards	leftwards (always opposite to moving direction)
zero	leftwards
leftwards	leftwards

11 Newton's third law of motion

To every action, there is an equal and opposite reaction. The action and reaction forces act on different interacting objects simultaneously.



Key terms

1 action-and-reaction pair 作用力—反作用力對 p.138
2 force 力 p.94
3 free-body diagram 隔離體圖 / 自由體圖 p.98
4 friction 摩擦力 p.96
5 inertia 慣性 p.102
6 kilogram (kg) 千克 / 公斤 p.104
7 mass 質量 p.104
8 net force 淨力 p.99
9 newton (N) 牛頓 p.95
10 Newton's first law of motion 牛頓運動第一定律 p.105
11 Newton's second law of motion 牛頓運動第二定律 p.109
12 Newton's third law of motion 牛頓運動第三定律 p.135
13 normal force 法向力 p.96
14 tension 張力 p.97
15 terminal speed 終端速率 p.126
16 weight 重量 p.97

Revision 3

Multiple-choice questions

*1 3.1 Which of the following is/are vector(s)?

(1) Weight

(2) Acceleration

(3) Speed

A (2) only

B (1) and (2) only

C (1) and (3) only

D (1), (2) and (3)

*2 3.5 Two objects fall freely from the same position in a vacuum. Which of the following statements must be correct?

(1) The objects have the same acceleration.

(2) The net forces acting on the objects are always the same.

(3) The net force acting on each object keeps decreasing.

A (1) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

*3 3.7 A boy is pushing a wall (Fig a).

Fig a

Which of the following is an action-and-reaction pair?

A The weight of the boy and the normal force acting on him by the ground

B The pushing force of the boy on the wall and the force acting on him by the wall

C The friction acting on the boy's feet and the force acting on him by the wall

D The friction acting on the boy's feet and his pushing force on the wall

*4 3.2 A tug boat towing a car ferry moves at a constant velocity (Fig b).

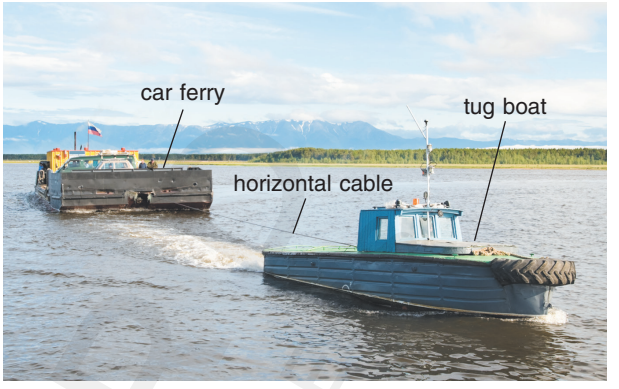


Fig b

What is the relationship between the tension T in the horizontal cable and the resistive force f acting on the car ferry by the water?

A T is smaller than f.

B T is as large as f.

C T is larger than f.

D The relationship cannot be determined.

*5 Misc Why should we hold the handrail when using an escalator (Fig c)?

Fig c

(1) A force from the handrail is needed to keep us moving with the escalator.

(2) The friction acting on us opposes our motion.

(3) We can stop ourselves in case the escalator stops suddenly.

A (2) only

B (3) only

C (1) and (2) only

D (1), (2) and (3)

★6 3.4 A tiger of mass 300 kg climbs up a vertical tree (Fig d).

Fig d

What is its acceleration when it applies a downward force of 3100 N on the tree?

A 0.523 m s^{-2} (upwards)
B 0.523 m s^{-2} (downwards)
C 10.3 m s^{-2} (upwards)
D 10.3 m s^{-2} (downwards)

★7 3.3 A boy kicks a ball of mass 1.2 kg on level grassland. The ball rolls for 20 m before it stops. Assume that the ball slows down uniformly. If the velocity of the ball after being kicked is u m s^{-1}, what is the net force acting on the ball when it rolls?

A $0.01u^2$ N
B $0.02u^2$ N
C $0.03u^2$ N
D $0.06u^2$ N

★8 Misc Which of the following is/are the direct implication(s) of inertia?

(1) A boy standing on a bus is 'pulled' backwards when the bus starts moving.
(2) If you push another person in front of you on a smooth surface, you will move backwards.
(3) The balls of different masses reach the ground at the same time in Galileo's experiment (carried out on the tower of Pisa).

A (1) only
B (2) only
C (3) only
D (1) and (2) only

★9 3.3 Figure e shows the relationship between the acceleration a and the mass m of an object when the object is acted on by a constant net force F. Find F.

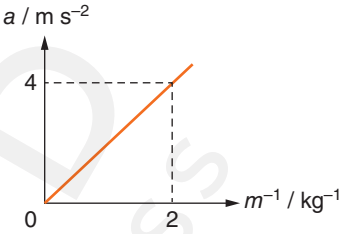


Fig e

A 0.5 N B 2 N
C 4 N D 8 N

★10 3.3 X and Y travel along the same straight road. Figure f shows their v–t graphs.

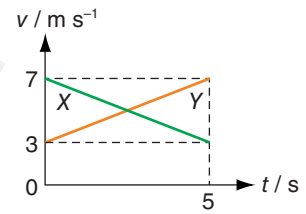


Fig f

Which of the following statements must be correct?

(1) X and Y move in opposite directions.
(2) The net forces acting on X and Y point in opposite directions.
(3) The net forces acting on X and Y have the same magnitude.

A (1) only B (2) only
C (2) and (3) only D (1), (2) and (3)

★11 3.6 A constant horizontal force of 20 N pulls an object of mass 2 kg along a straight line on a rough horizontal plane. Figure g shows the v–t graph of the object.

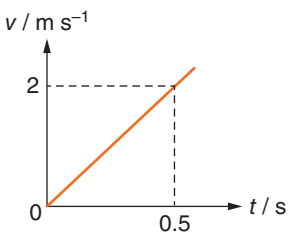


Fig g

What is the friction acting on the object?

A 8 N B 12 N
C 16 N D 28 N

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(For Q12–13.) An object is put on a rough horizontal surface. A horizontal force F acts on the object. Figure h shows how the acceleration a of the object changes with F. The friction acting on the object is constant when the object is in motion.

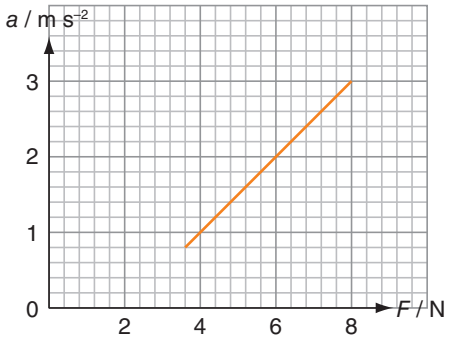


Fig h

★12 3.6 Estimate the mass of the object.

A 0.33 kg B 0.5 kg
C 2 kg D 3 kg

★13 3.6 Estimate the friction acting on the object.

A 2 N B 4 N
C 6 N D 8 N

★14 Misc Jeff pushes a wall with a force of 20 N and Ken pushes Jeff with a force of 15 N (Fig i). They both remain stationary.

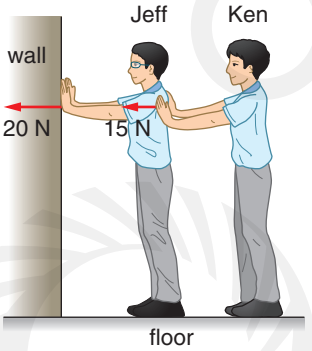


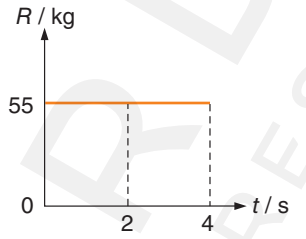
Fig i

Find the magnitude of the force F acting on Jeff by the wall and the friction f acting on Jeff by the floor.

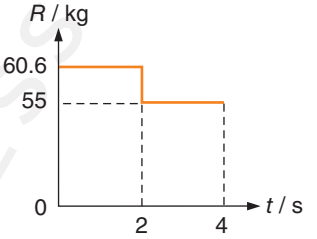
	F / N	f / N
A	5	15
B	20	5
C	20	15
D	35	20

★15 3.4 A woman of mass 55 kg is standing on a bathroom scale calibrated in kilograms inside a lift. The lift accelerates upwards at 1 m s^{-2} from rest from time t = 0 to t = 2 s. Then it moves with a constant velocity from t = 2 s to t = 4 s. Which of the following graphs best shows the variation in the reading R of the scale with t?

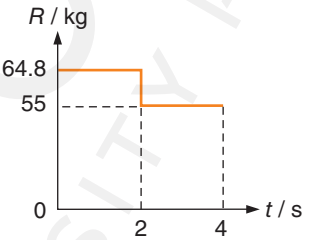
A



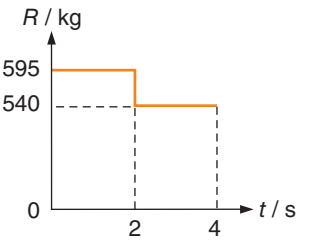
B



C



D



★★16 3.4 Amy is inside a lift which moves upwards. Figure j shows the v–t graph of the lift within a certain period of time.

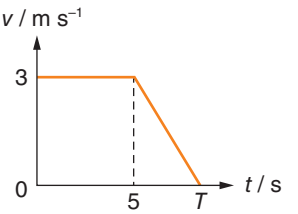


Fig j

The normal force acting on Amy by the lift floor is 500 N from t = 0 to t = 5 s and is 400 N from t = 5 s to t = T. Find T.

A 5.55 s
B 6.53 s
C 7.34 s
D 10.2 s

Refer p.123

Q19 Exam Report: Only one quarter of the candidates identified the correct answer as Option A, which suggests that a majority of candidates did not realize that friction applies as long as the block is still moving.

★★**17** 3.7 A man of weight W paints the ceiling with a paint roller of weight w (Fig k).

Fig k

At a certain instant, he holds the roller vertically and exerts an upward force F on it to press it against the ceiling. He and the roller remain stationary. What is the normal force acting on the man by the ground?

A $W - w + F$ B $W + w + F$

C $W - F$ D $W + F$

Refer p.139

18 3.6 **HKDSE 2012 Paper 1A Q11**

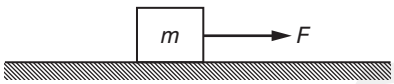
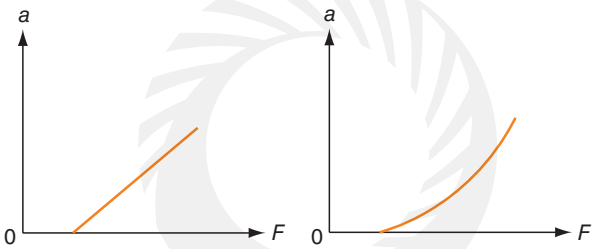


Fig l

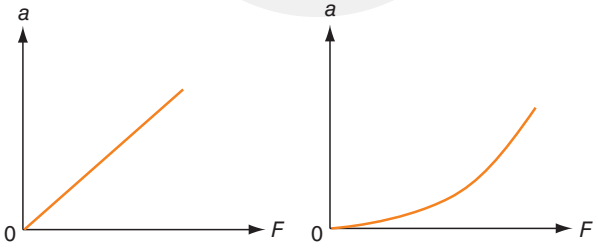
A block of mass m initially resting on a rough horizontal surface is pulled along the surface by a horizontal force F increasing from zero. If the frictional force is constant, which graph shows the relation between the acceleration of the block a and force F?

A (58%)



B

C



D

3.6 **19** **HKDSE 2013 Paper 1A Q7**

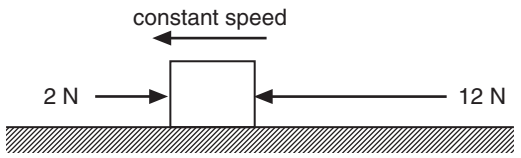


Fig m

A block on a rough horizontal surface is moving to the left with constant speed under two horizontal forces 2 N and 12 N indicated as shown. If the force of 12 N is suddenly removed, what is the net force acting on the block at that instant?

A 12 N (25%) B 10 N (15%)

C 8 N (26%) D 2 N (34%)

3.3 **20** **HKDSE 2015 Paper 1A Q5**

A constant net force acting on an object of mass m_1 produces an acceleration a_1 while the same force acting on another object of mass m_2 produces an acceleration a_2. If this net force acts on an object of mass $(m_1 + m_2)$, what would be the acceleration produced?

A $a_1 + a_2$ B $\dfrac{a_1 + a_2}{2}$

C $\dfrac{a_1 a_2}{a_1 + a_2}$ (64%) D $\dfrac{2a_1 a_2}{a_1 + a_2}$

3.7 **21** **HKDSE 2016 Paper 1A Q6**

A boy of weight W exerts a downward pulling force F on a rope of weight G hung vertically from the ceiling. He stands still on the ground as shown.

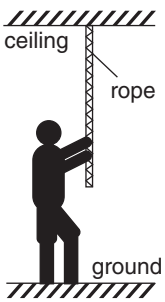


Fig n

Which of the following gives the magnitude of the force exerted by

(1) the boy on the ground;

(2) the rope on the ceiling?

	(1)	(2)
A	W	$G - F$
B	W	$G + F$
C	$W - F$	$G - F$
D	$W - F$	$G + F$ (57%)

Q22 Exam Report: Over half of the candidates wrongly thought that *R* and *W* is an action-and-reaction pair.

3.7 **22 HKDSE 2021 Paper 1A Q4**

A block of weight *W* is at rest on a horizontal ground as shown.

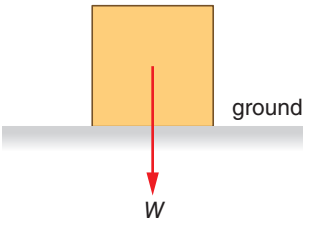


Fig o

The force acting on the block by the ground is *R*. Which of the following statements is/are correct?

(1) *R* and *W* are opposite in direction.

(2) *R* and *W* are equal in magnitude.

(3) *R* and *W* is an action-and-reaction pair.

A (1) only (3%)

B (1) and (2) only (43%)

C (2) and (3) only (1%)

D (1), (2) and (3) (53%)

Conventional questions

★**23** Misc Magnet *Y* is at rest on a horizontal table. Magnet *X* floats above *Y* as shown (Fig p). The two forces acting on *X* are its weight and the magnetic force. The weight of *X* is 2 N. The force between *X* and the pencil is negligible.

Fig p

(a) Draw the free-body diagram for *X*. (2 marks)

(b) Find the magnetic force acting on *X*. (2 marks)

(c) Name an action-and-reaction pair in this situation. (1 mark)

(d) Draw the free-body diagram for *Y*. (2 marks)

(e) The weight of *Y* is 2 N. Find the normal force acting on it by the table. (2 marks)

★**24** 3.3 Figure q shows how the net force *F* acting on an object varies with time *t*. The object is at rest at $t = 0$ and moves along a straight line for the entire journey. The mass of the object is 5 kg.

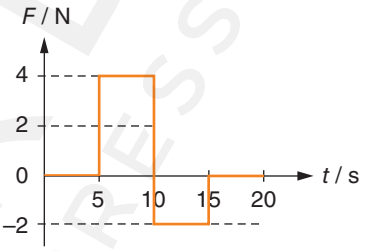


Fig q

(a) Describe the motion of the object from $t = 0$ to $t = 20$ s. (4 marks)

(b) Find the velocity of the object at $t = 20$ s. (3 marks)

★**25** Misc A baby is resting on a weighing scale (Fig r).

Fig r

Show that the force that the baby exerts on the scale is equal to the baby's weight. (4 marks)

★**26** 3.6 A 10-kg block is initially at rest on a horizontal rough surface. It is being pushed to the right by a 10-N force. The maximum friction between the block and the surface is 5 N.

(a) Draw the free-body diagram for the block. (2 marks)

(b) Find the net force acting on the block. (2 marks)

(c) Find the acceleration of the block. (2 marks)

(d) Find the displacement of the block after it has moved for 12 s. (2 marks)

★**27** Misc Wrestler *A* pushes her opponent *B* backwards.

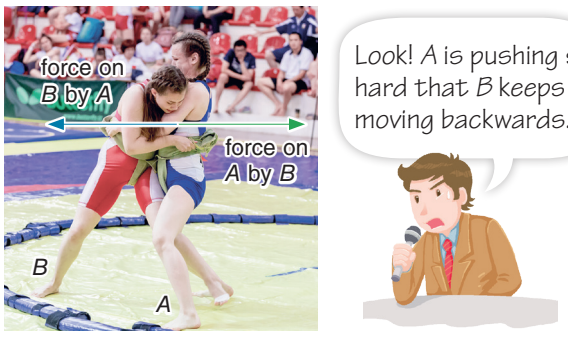


Fig s

(a) Who exerts a greater force on her opponent? (1 mark)

(b) Explain why *B* is pushed backwards. (2 marks)

★**28** 3.4 Sam is standing in a lift. The lift moves downwards and has a constant acceleration. The initial speed of the lift is 8 m s^{-1} and it comes to rest after 5 s. The mass of Sam is 60 kg.

(a) Find the acceleration of Sam during the deceleration. (2 marks)

(b) Find the apparent weight of Sam during the deceleration. (2 marks)

★**29** 3.6 A box which is initially at rest on a rough horizontal plane is pulled to the right by a constant force *F* from time $t = 0$ to $t = 2$ s (Fig t). *F* disappears after $t = 2$ s. Figure u shows the v–t graph of the box, with the direction towards the right taken as positive. The mass of the box is 0.5 kg. The friction acting on the box is constant when the box is moving.

Fig t

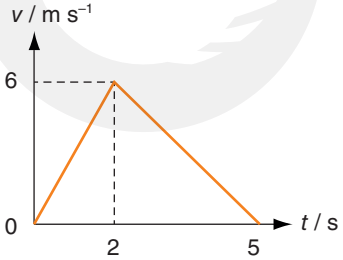


Fig u

(a) Find the acceleration of the box

(i) from 0–2 s, (1 mark)

(ii) from 2–5 s. (1 mark)

(b) Find the friction acting on the box when the box is moving. (2 marks)

(c) Find *F*. (2 marks)

★**30** Misc In an air hockey game, air is blown out from the holes on the table to produce a cushion of air so that the puck can move on it with nearly no friction (Fig v).

Fig v

The puck of mass 40 g is pushed by a force *F* for 0.2 s from rest. It continues to move after the force disappears. At $t = 0.4$ s, the table is turned off and stops blowing out air. Figure w shows the v–t graph of the puck from $t = 0$ to $t = 0.8$ s. It moves along a straight line in this period of time.

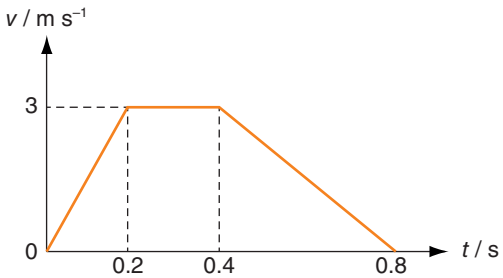


Fig w

(a) Describe the motion of the puck from $t = 0$ to $t = 0.8$ s. (3 marks)

(b) Explain the motion of the puck from $t = 0.4$ s to $t = 0.8$ s. (2 marks)

(c) Find the distance travelled by the puck from $t = 0$ to $t = 0.8$ s. (2 marks)

(d) Find *F*. (2 marks)

(e) Suppose the puck is pushed by a larger constant force from $t = 0$ to $t = 0.2$ s. Sketch the new v–t graph in Figure w. (2 marks)

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*31 3.3 A 4-kg box is initially at rest on a smooth horizontal plane. It is pulled by a constant 2-N force towards the right from time $t = 0$ to $t = 3$ s (Fig x).

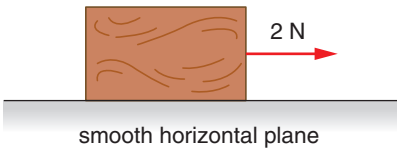

Fig x

(a) Draw the free-body diagram for the box. (2 marks)

(b) Find its acceleration. (2 marks)

(c) Find its displacement from $t = 0$ to $t = 3$ s. (2 marks)

(d) Find its velocity at $t = 3$ s. (2 marks)

(e) The 2-N force disappears after $t = 3$ s. Sketch the s–t and v–t graphs of the box from $t = 0$ to $t = 5$ s. (4 marks)

*32 Misc A man takes a lift to go up from the ground floor (Fig y).

Fig y

(a) Draw the free-body diagrams for the man and the lift separately. (4 marks)

(b) Write down an action-and-reaction pair in this situation. (1 mark)

(c) The masses of the man and the lift are 65 kg and 900 kg respectively. What is the normal force acting on the man by the lift floor when the lift accelerates upwards at 0.6 m s^{-2}? (2 marks)

*33 3.4 Read the following passage about **drop tower** and answer the questions that follow.

Drop towers are commonly seen in amusement parks (Fig z). At the beginning of the ride, the seats carrying the passengers are lifted from the ground to the top of the tower. After staying at rest for a few seconds, they will fall freely. Finally, brakes are activated to slow down the seats to a stop.

The v–t graph of a passenger on the ride is approximated in Figure aa, with upwards taken as positive.

Fig z

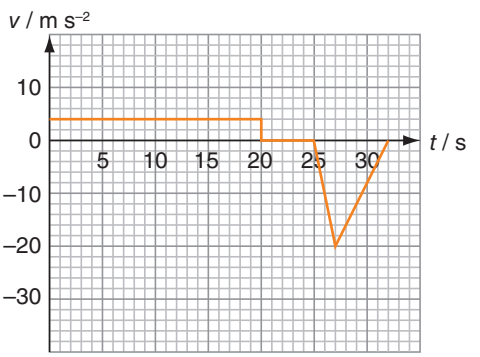

Fig aa

Assume that a passenger of 65 kg moves as the v–t graph in Figure aa.

(a) Find the force exerted by the seat on the passenger when the seat moves from the ground to the top of the tower. (1 mark)

(b) Draw a free-body diagram for the passenger when he is accelerating downwards. (1 mark)

(c) When is the force exerted by the seat on the passenger the largest? Find its magnitude. (3 marks)

Q36 Exam Report: This was a typical question on kinematics involving apparent weight. Candidates' performance was satisfactory. Parts (a) and (b) were well answered. In (c)(i), some candidates failed to verify the result as they made mistakes in the sign of acceleration in the calculation. Most were able to obtain the answer in (c)(ii).

★★**34** 3.5 Jackie dives from 6000 m above the ground (Fig ab) and opens her parachute when she is at a height of 2000 m. Assume that she falls vertically from rest and reaches the terminal speed before opening the parachute.

Fig ab

Consider the time period from the instant she leaves the aeroplane to the instant she opens her parachute.

(a) Draw a free-body diagram for Jackie. (2 marks)

(b) How will the forces in (a) change during the fall? (3 marks)

(c) Describe Jackie's motion during the fall. (3 marks)

(d) (i) Take downwards as positive. Sketch the v–t graph of Jackie. (2 marks)

(ii) What is the area under the v–t graph? (1 mark)

Refer p.126

★★**35** 3.7 Figure ac shows a drone hovering in mid-air in a stable position. It contains powerful fans that blow air downwards.

Fig ac

(a) Draw the free-body diagram for the drone. (2 marks)

(b) Explain why the drone is able to hover in mid-air. (3 marks)

(c) If the drone hovers at a few centimetres above a weighing scale, is the reading of the scale equal to zero? Explain your answer. (3 marks)

Refer p.136

3.4 **36** **HKDSE 2016 Paper 1B Q3**

A person of mass m stands on a balance inside a lift. The lift goes down from the top of a building at time $t = 0$ and it reaches the ground at $t = T$. The velocity–time (v–t) graph of the lift is shown in Figure ad.

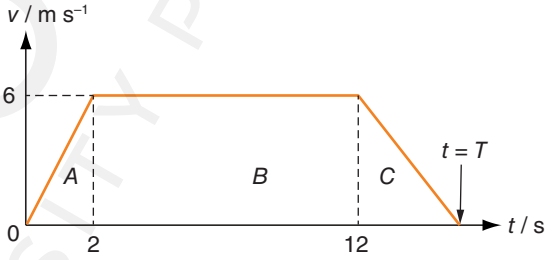

Fig ad

(a) Calculate the acceleration of the lift from $t = 0$ to $t = 2$ s. (2 marks)

The reading of the balance changes during the person's ride on the lift and the readings registered are 685 N, 569 N and 395 N.

(b) Match these readings with the three stages, A, B and C, of the ride (shown in Figure ad). Hence deduce the mass of the person. (3 marks)

(c) (i) Show that $T = 15$ s. (2 marks)

(ii) Hence estimate the height of the building. (2 marks)

3.4 **37** **HKDSE 2017 Paper 1B Q4(b)**

If a small ball is released from rest from the top of a cliff, the speed of the ball becomes constant after a period of time. By considering the forces acting on the ball and using Newton's laws of motion, explain why the speed of the ball becomes constant. (3 marks)

Q37 Exam Report: This part revealed that quite a number of the candidates had misconceptions about Newton's third law of motion — taking weight and air resistance as an action-and-reaction pair in this case.

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Know more

Aristotle and Galileo (p.101)

The idea that a force is needed to keep an object moving was put forward over 2000 years ago by the Greek philosopher Aristotle.

Towards the end of the 16th century, Galileo questioned this idea. He proposed a thought experiment: a ball was released from one end of a frictionless bent rail, the other end of which could be tilted to different angles (Fig a).

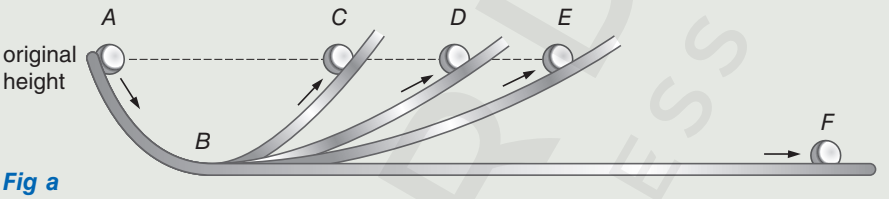


Fig a

He argued that the ball would rise to the same height as the point of release *A* when the rail was tilted to different angles. If the rail was bent along *ABF* where *BF* was horizontal, the ball would never rise up to the same height as *A* and it would move forever at a constant speed along a straight line.

Galileo just constructed the experiment in his mind and argued the results. Therefore, this experiment is called a thought experiment. He based his argument on the pin-and-pendulum experiment that he actually conducted (Fig b).

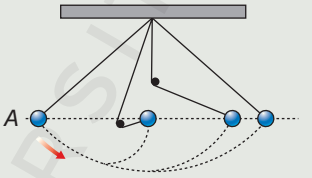
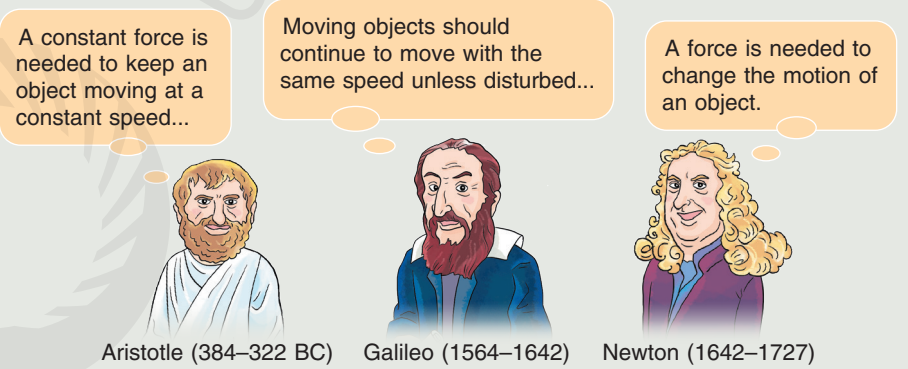


Fig b

He found that the bob rose to the same height as the initial position *A*, no matter which path it took. This observation supported the argument of his thought experiment.

Know more

Development of the idea of force and motion



Newton made a rather accurate description of the relationship between force and motion. But subsequently it was discovered that Newton's laws fail at extremely high speeds (near the speed of light). *Albert Einstein* (1879–1955) developed the theory of *special relativity*, which gives a more accurate description of the relationship between force and motion at extremely high speeds.

Today many scientists are still working hard to refine the theories and laws of physics.

Albert Einstein 愛因斯坦　　special relativity 狹義相對論